Solve for x.

$2(3x - 1) - (x + 4) = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q1/5

801

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{2}(4x + 6) = 3x - 5$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q2/5

802

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q2/5

Solve for x.

$0.5(x + 8) - 2x = 1$

A. $x = 6$

B. $x = 3$

C. $x = 2$

D. $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q3/5

803

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q3/5

True or false?

$x = 5$ is the solution of $3(x - 2) = 2x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q4/5

804

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q4/5

True or false?

$x = 1$ is the solution of $4x - (x + 2) = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q5/5

805

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 2

Interpret the terms, factors, and coefficients of an expression • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

What is the coefficient of x in $5x^2 - 3x + 8$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q1/5

807

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q1/5

How many terms are in $4a^3 - 2a + 7$?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q2/5

808

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q2/5

Which is a factor of $7(x + 2)^2$?

A. x^2

B. $(x + 2)^2$

C. 14

D. 7x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q3/5

809

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x + 2)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q3/5

True or false?

In −6y + 9, the constant term is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q4/5

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q4/5

True or false?

In $3x^2 + x$, the coefficient of x is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q5/5

811

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The coefficient of x is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Factor and expand linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Identify the parts of an expression	**6.EE.A.2.b**
FORWARD 1	Expand to decide how many solutions an equation has	**8.EE.C.7.a**
FORWARD 2	Factor quadratic expressions	**HSA-SSE.A.2**

SET 27 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$-4(2x - 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q1/5

814

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-8x + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q1/5

Factor out the greatest common factor.

6y + 15

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q2/5

815

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

3(2y + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q2/5

Which expression is equivalent to $\frac{1}{2}(8a - 6)$?

A. $8a - 3$

B. $4a - 3$

C. $4a - 6$

D. $16a - 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q3/5

816

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4a − 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q3/5

True or false?

$12m - 18 = 6(2m - 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q4/5

817

QUESTION ID

S27-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q4/5

True or false?

$-5(n + 2) = -5n + 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q5/5

818

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-5(n + 2) = -5n - 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q1/5

820

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q1/5

Use the greatest common factor to write 30 + 42 as a product.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q2/5

821

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6(5 + 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q2/5

Which expression is equal to 18 + 27?

A. 9(2 + 3)

B. 9(2 + 27)

C. 9(18 + 3)

D. 3(6 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q3/5

822

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9(2 + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q3/5

True or false?

The greatest common factor of 16 and 40 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q4/5

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q4/5

True or false?

$20 + 35 = 5(4 + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q5/5

824

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $20 + 35 = 5(4 + 7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 2

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms are in 3x + 7y − 2?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q1/5

826

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q1/5

What is the coefficient of w in 9 + 5w?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q2/5

827

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q2/5

Which part of 4(n + 6) is a sum?

A. 4n

B. 4

C. n + 6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q3/5

828

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

In 2(a + b), the factor 2 multiplies the sum a + b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q4/5

829

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

In 8k − 3, the term 8k is a constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q5/5

830

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 7.EE.A.1

FORWARD 1

Expand to decide how many solutions an equation has • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x - 2) = 4x - 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q1/5

832

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$-3(x + 1) = -3x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q2/5

833

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q2/5

Which equation has no solution?

A. 2(x + 5) = x + 10

B. 2(x + 5) = 3x

C. 2(x + 5) = 2x + 10

D. 2(x + 5) = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q3/5

834

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2(x + 5) = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q3/5

True or false?

$6(x - 1) = 2(3x - 3)$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q4/5

835

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q4/5

True or false?

5(x + 1) = 5x + 1 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q5/5

836

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q5/5

SET 27 • 7.EE.A.1

FORWARD 2

Factor quadratic expressions • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 7x + 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q1/5

838

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q1/5

Factor completely.

$3x^2 - 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q2/5

839

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3(x + 2)(x − 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q2/5

Factor.

$x^2 - 2x - 15$

A. $(x - 5)(x - 3)$

B. $(x - 15)(x + 1)$

C. $(x + 5)(x - 3)$

D. $(x - 5)(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q3/5

840

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x − 5)(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

$2x^2 + 10x = 2x(x + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q4/5

841

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q5/5

842

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 5x + 6 = (x + 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 7.EE.A.2 • EXPRESSIONS & EQUATIONS

Rewrite an expression to show how the quantities in a problem are related

MAIN	Grade 7 main question	**7.EE.A.2**
BACKWARD 1	Write expressions from words	**6.EE.A.2.a**
BACKWARD 2	Write and interpret numerical expressions	**5.OA.A.2**
FORWARD 1	Write equivalent expressions with the properties of integer expon…	**8.EE.A.1**
FORWARD 2	Choose an equivalent form of an expression to reveal a property	**HSA-SSE.B.3**

SET 28 • 7.EE.A.2

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.2

A shirt costs p dollars. The price is raised by 30%.

Write the new price as a single term.

New price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q1/5

845

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

1.3p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle's length is 4 cm more than its width w.

Write its perimeter in two equivalent ways.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q2/5

846

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

2w + 2(w + 4) and 4w + 8

GRADING NOTE Any two correct equivalent expressions earn credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q2/5

A price p is lowered by 15%.

Which expression shows the new price?

A. 1.15p

B. 0.85p

C. 0.15p

D. p − 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q3/5

847

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.85p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q3/5

True or false?

n + 0.08n = 1.08n shows that increasing n by 8% is the same as multiplying n by 1.08.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q4/5

848

QUESTION ID

S28-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q4/5

True or false?

The price after a 20% discount on a price of d dollars is d − 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q5/5

849

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is 0.8d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 1

Write expressions from words • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "5 less than y."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q1/5

851

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q1/5

Write an expression for "the product of 8 and n."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q2/5

852

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q2/5

Which expression means "the sum of x and 3, times 4"?

A. 4(x + 3)

B. x + 3 × 4

C. 4x + 3

D. 4 + x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q3/5

853

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q3/5

True or false?

"Twice a number m" can be written as 2m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q4/5

854

QUESTION ID

S28-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q4/5

True or false?

"7 fewer than k" can be written as 7 − k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q5/5

855

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is k − 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 2

Write and interpret numerical expressions • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 6 and 9, then multiply by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q1/5

857

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 × (6 + 9)

GRADING NOTE (6 + 9) × 2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q1/5

How does 5 × (24 + 13) compare with 24 + 13?

A. It is 5 more.

B. It is 5 less.

C. It is 5 times as large.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q2/5

858

QUESTION ID

S28-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. It is 5 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q2/5

Write an expression for "subtract 4 from 20, then divide by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q3/5

859

QUESTION ID

S28-B2-Q3

TYPE

Short answer

CORRECT ANSWER

(20 − 4) ÷ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q3/5

True or false?

(7 + 8) × 3 is three times as large as 7 + 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q4/5

860

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q4/5

True or false?

$10 \times (5 - 2)$ is 10 more than $5 - 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q5/5

861

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 1

Write equivalent expressions with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$4^{5} \times 4^{-2}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

863

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4^{3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

Write as a single power.

$(2^3)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2^{12}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

Which number is equal to $5^6 \div 5^8$?

A. 25

B. 5^{14}

C. $\frac{1}{25}$

D. -25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

True or false?

$7^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

866

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

True or false?

$3^2 \times 3^3 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

867

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 \times 3^3 = 3^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 2

Choose an equivalent form of an expression to reveal a property • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3

Factor $f(x) = x^2 - 6x + 8$ to find the values of x where $f(x) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q1/5

869

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q1/5

A price p is raised 10%, and then the new price is raised 10% again: p(1.1)(1.1).

Write this as a single factor times p, and give the total percent increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q2/5

870

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.21p; a 21% increase

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q2/5

A value after t years is 50(1.03)^t.

What is the yearly percent increase?

A. 50%

B. 3%

C. 0.3%

D. 103%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q3/5

871

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q3/5

True or false?

Writing $x^2 - 9$ as $(x - 3)(x + 3)$ shows that the expression equals 0 when $x = 3$ or $x = -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q4/5

872

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q4/5

True or false?

A 10% increase followed by a 10% decrease leaves a price unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q5/5

873

QUESTION ID

S28-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The final price is 0.99 times the original price.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Solve multistep problems with positive and negative rational numbers in any form

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Interpret a fraction as division	**5.NF.B.3**
BACKWARD 2	Find a percent of a quantity	**6.RP.A.3.c**
FORWARD 1	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 2	Represent constraints with equations or inequalities	**HSA-CED.A.3**

SET 29 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Jon earns $16.50 per hour. He gets a raise of $\frac{1}{5}$ of his wage. Then he works 30 hours.

How much does he earn for the 30 hours?

Earnings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q1/5

876

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

$594

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A board is $8\frac{1}{4}$ feet long. Ana cuts off 2.5 feet. She cuts the rest into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q2/5

877

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

1 11/12 feet

GRADING NOTE 23/12 feet, or about 1.92 feet, is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q2/5

A 60-gallon tank is $\frac{3}{4}$ full. Then 12.5 gallons are used.

How many gallons are left?

A. 57.5

B. 47.5

C. 32.5

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q3/5

878

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q3/5

True or false?

A $40 sweater is 25% off. Mia pays with a $50 bill. Her change is $20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q4/5

879

QUESTION ID

S29-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q4/5

True or false?

Leo runs $2\frac{1}{2}$ miles on Monday and 3.4 miles on Tuesday. He runs 6.4 miles in all.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q5/5

880

QUESTION ID
S29-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE He runs 5.9 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 1

Interpret a fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $\frac{3}{4}$ as a division expression. Then find its decimal value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q1/5

882

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3 ÷ 4 = 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q1/5

5 friends share 2 pizzas equally.

How much pizza does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q2/5

883

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2/5 of a pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q2/5

Which number is equal to 7 ÷ 8?

A. 7.8

B. $\frac{7}{8}$

C. $\frac{8}{7}$

D. 0.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q3/5

884

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{2} = 9 \div 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q4/5

885

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q4/5

True or false?

$3 \div 5 = \frac{5}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q5/5

886

QUESTION ID

S29-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 2

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 35% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q1/5

888

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q1/5

What is 12.5% of 64?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q2/5

889

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 150% of 20?

A. 130

B. 30

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q3/5

890

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q3/5

True or false?

60% of 25 is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q4/5

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q4/5

True or false?

5% of 300 is 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q5/5

892

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5% of 300 is 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 1

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

Adult tickets cost $9 and child tickets cost $5. Someone buys 20 tickets for $140.

How many adult tickets were bought?

Adult tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q1/5

894

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q1/5

Plan A costs $20 plus $0.10 per minute. Plan B costs $0.35 per minute.

For how many minutes do the plans cost the same?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q2/5

895

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q2/5

Two numbers have a sum of 30 and a difference of 8.

What is the larger number?

A. 15

B. 11

C. 19

D. 22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q3/5

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q3/5

True or false?

$x = 3$ and $y = 5$ is a solution of both $x + y = 8$ and $2x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q4/5

897

QUESTION ID

S29-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q4/5

True or false?

$x = 4$ and $y = 2$ is a solution of both $x + y = 6$ and $x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q5/5

898

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 2

Represent constraints with equations or inequalities • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Write an inequality for m muffins and l loaves.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q1/5

900

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$0.75m + 2.5l \leq 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q1/5

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Can she make 12 muffins and 7 loaves? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q2/5

901

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. 12 muffins and 7 loaves need 9 + 17.5 = 26.5 cups, which is more than 24 cups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q2/5

A trip budget is at most $150. Gas costs $3.50 per gallon (g) and meals cost $12 each (f).

Which inequality shows the budget?

A. $3.5g + 12f \leq 150$

B. $12g + 3.5f \leq 150$

C. $3.5 + 12 \leq 150gf$

D. $3.5g + 12f \geq 150$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q3/5

902

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3.5g + 12f ≤ 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q3/5

True or false?

m = 8 and l = 4 makes 0.75m + 2.5l ≤ 24 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q4/5

903

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q4/5

True or false?

m = 10 and l = 9 makes 0.75m + 2.5l ≤ 24 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q5/5

904

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.75(10) + 2.5(9) = 30, which is more than 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Use estimation to judge whether an answer is reasonable

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Round decimals	**5.NBT.A.4**
BACKWARD 2	Use rounding to check whether an answer is reasonable	**4.OA.A.3**
FORWARD 1	Estimate very large or very small quantities with powers of 10	**8.EE.A.3**
FORWARD 2	Define appropriate quantities and units for a model	**HSN-Q.A.2**

SET 30 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate 19.8 × 4.1 by rounding each number to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q1/5

907

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

80

GRADING NOTE 20 × 4 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q1/5

Kim says 48.7 + 31.2 = 799.

Use estimation to explain whether her answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q2/5

908

QUESTION ID

S30-M-Q2

TYPE

Short answer

CORRECT ANSWER

Not reasonable. 49 + 31 is about 80, so the sum should be about 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q2/5

A meal costs $38.75. Ana adds a 20% tip.

Which is the best estimate of the total?

A. About $40

B. About $58

C. About $78

D. About $47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q3/5

909

QUESTION ID

S30-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. About $47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q3/5

True or false?

$6\frac{7}{8} \times 3\frac{1}{10}$ is about 7×3, so a product of about 21 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q4/5

910

QUESTION ID

S30-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q4/5

True or false?

A $59.99 jacket is 30% off. A sale price of about $18 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q5/5

911

QUESTION ID

S30-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sale price should be about $42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 1

Round decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 7.86 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q1/5

913

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q1/5

Round 12.348 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q2/5

914

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q2/5

Round 4.165 to the nearest hundredth.

A. 4.17

B. 4.1

C. 4.2

D. 4.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q3/5

915

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q3/5

True or false?

3.97 rounded to the nearest tenth is 4.0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q4/5

916

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q4/5

True or false?

15.52 rounded to the nearest whole number is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q5/5

917

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 2

Use rounding to check whether an answer is reasonable • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

Estimate 398 + 211 by rounding each number to the nearest hundred.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q1/5

919

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q1/5

Lia says 52 × 4 = 2,008.

Round 52 to the nearest ten to decide whether her answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q2/5

920

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Not reasonable.
50 × 4 = 200, so the answer should be about 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q2/5

Round each number to the nearest hundred.

What is the best estimate of 689 − 312?

A. 300

B. 400

C. 500

D. 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q3/5

921

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q3/5

True or false?

Rounding 49 to the nearest ten gives an estimate of 300 for 49 × 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q4/5

922

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q4/5

True or false?

Rounding each number to the nearest hundred gives an estimate of 1,200 for 812 − 397.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q5/5

923

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The estimate is 800 − 400 = 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 1

Estimate very large or very small quantities with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Estimate 38,900,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q1/5

925

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q1/5

Estimate 0.00062 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q2/5

926

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6×10^{-4}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q2/5

About how many times as large as 3×10^5 is 9×10^8?

A. 300

B. 3

C. 30

D. 3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q3/5

927

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q3/5

True or false?

3×10^8 is a reasonable estimate for 296,000,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q4/5

928

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} is about 6,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q5/5

929

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $6 \times 10^{-3} = 0.006$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 2

Define appropriate quantities and units for a model • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2

A heart beats about 70 times per minute.

About how many times does it beat in 1 day? Round to the nearest ten thousand.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q1/5

931

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

100,000

GRADING NOTE 70 × 1,440 = 100,800.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q1/5

Which unit best describes the distance between two cities: millimeters, meters, or kilometers?

Unit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q2/5

932

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q2/5

Which quantity is most useful for estimating the cost to paint the walls of a room?

A. The volume of the room

B. The height of the door

C. The number of windows

D. The area of the walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q3/5

933

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The area of the walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

Gallons of water used per day is a reasonable quantity for modeling a family's water use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q4/5

934

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q4/5

True or false?

The color of a car is a useful quantity for modeling how much fuel it uses on a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q5/5

935

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form px + q = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve one-step equations	**6.EE.B.7**
BACKWARD 2	Write an expression with a variable for a situation	**6.EE.B.6**
FORWARD 1	Solve equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 31 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $25 sign-up fee plus $18 per month. Ty has paid $151 in all.

Write and solve an equation to find the number of months m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q1/5

938

QUESTION ID
S31-M-Q1

TYPE
Short answer

CORRECT ANSWER
18m + 25 = 151; m = 7

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q1/5

Solve for x.

$4x + 9 = -11$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q2/5

939

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q2/5

Ava has $60. She buys movie tickets for $7.50 each and has $15 left.

Which equation finds the number of tickets t?

A. $60t - 7.5 = 15$

B. $7.5 - 60t = 15$

C. $7.5t + 60 = 15$

D. $60 - 7.5t = 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q3/5

940

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60 − 7.5t = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q3/5

True or false?

$x = 6$ is the solution of $\frac{1}{2}x + 3 = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q4/5

941

QUESTION ID

S31-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q4/5

True or false?

A plumber charges $45 plus $30 per hour. A bill of $165 means the plumber worked 5 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q5/5

942

QUESTION ID

S31-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The plumber worked 4 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 1

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 13 = 41

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q1/5

944

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q1/5

Solve for y.

9y = 72

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q2/5

945

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q2/5

Solve for n.

$n - 6.4 = 3.6$

A. $n = 10$

B. $n = 2.8$

C. $n = -2.8$

D. $n = 23.04$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q3/5

946

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. n = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q3/5

True or false?

k = 7 is the solution of 5k = 35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q4/5

947

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q4/5

True or false?

p = 9 is the solution of p + 4 = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q5/5

948

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 2

Write an expression with a variable for a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

A ticket costs $12.

Write an expression for the cost of t tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q1/5

950

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q1/5

Ana has 5 more stickers than Ben. Ben has b stickers.

Write an expression for the number of stickers Ana has.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q2/5

951

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

b + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q2/5

A pizza has s slices. 4 friends share it equally.

Which expression gives the number of slices each friend gets?

A. $s - 4$

B. $4s$

C. $s + 4$

D. $\frac{s}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q3/5

952

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. s/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q3/5

True or false?

A pen costs p dollars. 3 pens cost 3p dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q4/5

953

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q4/5

True or false?

Mo is m years old. His age in 6 years is 6m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q5/5

954

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE His age in 6 years is m + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 1

Solve equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges $40 plus $5 per day. Store B charges $25 plus $8 per day.

After how many days do the stores charge the same amount?

Days: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q1/5

956

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q1/5

Solve for x.

$7x - 4 = 3x + 20$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q2/5

957

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q2/5

Solve for x.

$-2x + 9 = 4x - 15$

A. $x = -4$

B. $x = 1$

C. $x = 4$

D. $x = 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q3/5

958

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q3/5

True or false?

$x = -1$ is the solution of $5x + 3 = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q4/5

959

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q4/5

True or false?

$x = 2$ is the solution of $6x - 1 = 4x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q5/5

960

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 2

Rearrange a formula to solve for a chosen quantity • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2l + 2w for l.

l = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q1/5

962

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(P − 2w)/2

GRADING NOTE l = P/2 − w is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q1/5

Solve y = mx + b for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q2/5

963

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(y − b)/m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q2/5

Solve $A = \frac{1}{2}bh$ for h.

A. $h = 2Ab$

B. $h = \frac{A}{2b}$

C. $h = \frac{b}{2A}$

D. $h = \frac{2A}{b}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q3/5

964

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. h = 2A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q3/5

True or false?

Solving $d = rt$ for t gives $t = \frac{d}{r}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q4/5

965

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $C = 5n + 20$ for n gives $n = 5C - 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q5/5

966

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $n = (C - 20)/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form p(x + q) = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Evaluate expressions with parentheses	**5.OA.A.1**
BACKWARD 2	Use the distributive property to expand an expression	**6.EE.A.3**
FORWARD 1	Solve equations with the distributive property on both sides	**8.EE.C.7.b**
FORWARD 2	Solve quadratic equations by square roots or factoring	**HSA-REI.B.4.b**

SET 32 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle is 7 cm wide and (x + 3) cm long. Its area is 63 cm².

Write and solve an equation to find x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q1/5

969

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

7(x + 3) = 63; x = 6

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q1/5

Solve for x.

$-3(x - 4) = 27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q2/5

970

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q2/5

5 friends each buy a $4 snack and a drink that costs d dollars. They spend $32.50 in all.

Which equation finds d?

A. d + 5(4) = 32.5

B. 5(d + 4) = 32.5

C. 5d + 4 = 32.5

D. 4(d + 5) = 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q3/5

971

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5(d + 4) = 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q3/5

True or false?

$x = 2.5$ is the solution of $2(x + 1.5) = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q4/5

QUESTION ID

S32-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q4/5

True or false?

The solution of $\frac{1}{3}(x - 6) = 4$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q5/5

973

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q5/5

SET 32 • 7.EE.B.4.a

BACKWARD 1

Evaluate expressions with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$3 \times (8 + 4)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q1/5

975

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q1/5

Evaluate.

$(15 - 6) \times 2$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q2/5

976

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate.

$4 \times (9 - 2) + 1$

A. 25

B. 29

C. 36

D. 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q3/5

977

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

$2 \times (5 + 5) = 20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q4/5

978

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q4/5

True or false?

$(12 + 3) \times 2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q5/5

979

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

BACKWARD 2

Use the distributive property to expand an expression • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite using the distributive property.

9(m + 2)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q1/5

981

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9m + 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite using the distributive property.

4(3k + 5)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q2/5

982

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12k + 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q2/5

Which expression is equivalent to 6(x + 1.5)?

A. 6x + 1.5

B. 6x + 9

C. 6x + 7.5

D. 7x + 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q3/5

983

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q3/5

True or false?

$5(2 + y) = 10 + 5y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q4/5

984

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q4/5

True or false?

$8(p + 3) = 8p + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q5/5

985

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 1

Solve equations with the distributive property on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$3(x + 2) = 2(x + 7)$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q1/5

987

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q1/5

Solve for x.

$-2(x - 5) = 4(x + 1)$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q2/5

988

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$\frac{1}{2}(6x - 4) = 2(x + 3)$

A. x = 2

B. x = 8

C. x = −8

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q3/5

989

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q3/5

True or false?

$x = 4$ is the solution of $5(x - 1) = 3(x + 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q4/5

990

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $2(x + 6) = 4(x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q5/5

991

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 2

Solve quadratic equations by square roots or factoring • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve for x.

$(x + 3)^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q1/5

993

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 or −10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$x^2 = 81$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q2/5

994

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9 or −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q2/5

Solve for x.

$(x - 2)(x + 5) = 0$

A. $x = -5$ only

B. $x = 2$ only

C. $x = -2$ or $x = 5$

D. $x = 2$ or $x = -5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q3/5

995

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 or x = −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q3/5

True or false?

The solutions of $2(x - 1)^2 = 18$ are $x = 4$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q4/5

996

QUESTION ID

S32-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 = 16$ has only one solution, $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q5/5

997

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 4$ or $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Compare an algebraic solution with an arithmetic solution

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve two-step word problems	**3.OA.D.8**
BACKWARD 2	Decide whether a value makes an equation true	**6.EE.B.5**
FORWARD 1	Solve systems of two linear equations algebraically and by inspec…	**8.EE.C.8.b**
FORWARD 2	Explain each step in solving an equation	**HSA-REI.A.1**

SET 33 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Ron has $50. He buys 3 books that cost the same amount and has $14 left.

Arithmetic: (50 − 14) ÷ 3. Algebra: 3b + 14 = 50.

What is the price of one book?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q1/5

1000

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q1/5

An arithmetic solution to 4(x + 2) = 36 is 36 ÷ 4 − 2.

Describe the operations in order, and give x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q2/5

1001

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

Divide 36 by 4, then subtract 2; x = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which arithmetic steps solve 5x − 8 = 27?

A. Multiply 27 by 5, then add 8.

B. Add 8 to 27, then divide by 5.

C. Divide 27 by 5, then add 8.

D. Subtract 8 from 27, then divide by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q3/5

1002

QUESTION ID

S33-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Add 8 to 27, then divide by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q3/5

True or false?

Solving 2x + 6 = 20 with algebra and computing (20 − 6) ÷ 2 both give 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q4/5

1003

QUESTION ID

S33-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q4/5

True or false?

The arithmetic steps "subtract 4 from 15, then divide by 3" solve $3(x - 4) = 15$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q5/5

1004

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The correct steps are to divide 15 by 3, then add 4, so x = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q5/5

SET 33 • 7.EE.B.4.a

BACKWARD 1

Solve two-step word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

Jo has 45 stickers. She gives away 9. Then she shares the rest equally among 4 friends.

How many stickers does each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q1/5

1006

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

There are 6 boxes with 8 pens in each box. 5 pens are lost.

How many pens are left?

Pens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q2/5

1007

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q2/5

A book has 90 pages. Kim reads 20 pages on Monday and 35 pages on Tuesday.

How many pages are left?

A. 15

B. 55

C. 145

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q3/5

1008

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q3/5

True or false?

3 packs of 7 cards and 4 more cards make 25 cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q4/5

1009

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

40 apples are shared equally in 5 bags. Then 2 apples are taken out of each bag. Each bag now has 8 apples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q5/5

1010

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each bag has 6 apples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q5/5

1010

SET 33 • 7.EE.B.4.a

BACKWARD 2

Decide whether a value makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 4 a solution of 3x + 2 = 14? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q1/5

1012

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 3(4) + 2 = 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q1/5

Which value of n makes n + 9 = 15 true?

A. 9

B. 24

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q2/5

1013

QUESTION ID

S33-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which of the values 2, 5, and 8 makes 7y = 35 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q3/5

1014

QUESTION ID

S33-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q3/5

True or false?

m = 3 makes 12 ÷ m = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q4/5

1015

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q4/5

True or false?

a = 5 makes 2a − 1 = 11 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q5/5

1016

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 1

Solve systems of two linear equations algebraically and by inspection • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve the system.

$y = 2x$

$x + y = 18$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q1/5

1018

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 6, y = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q1/5

Solve the system.

$y = x + 3$

$y = -2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q2/5

1019

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2, y = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q2/5

Look at the system. How many solutions does it have?

$3x + 2y = 5$

$3x + 2y = 6$

A. Infinitely many solutions

B. Two solutions

C. No solution

D. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q3/5

1020

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q3/5

True or false?

The system y = 4x and y = 4x + 1 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q4/5

1021

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 4$ and $y = 6$ is the solution of the system $x + y = 10$ and $x - y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q5/5

1022

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 6$, $y = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 2

Explain each step in solving an equation • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

In solving an equation, $5(x - 2) = 30$ is rewritten as $5x - 10 = 30$.

Which property is used?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q1/5

1024

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The distributive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q1/5

In solving an equation, $5x - 10 = 30$ is rewritten as $5x = 40$.

Explain this step.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q2/5

1025

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 is added to both sides (addition property of equality).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q2/5

Which step changes 5x = 40 into x = 8?

A. Add 5 to both sides.

B. Subtract 5 from both sides.

C. Divide both sides by 5.

D. Multiply both sides by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q3/5

1026

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Divide both sides by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q3/5

True or false?

Each step in solving a linear equation should give an equation with the same solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q4/5

1027

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q4/5

True or false?

Subtracting 3 from only one side of an equation keeps the equation true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q5/5

1028

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Solve word problems that lead to inequalities of the form px + q > r or px + q < r

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Write an inequality for a situation	**6.EE.B.8**
BACKWARD 2	Decide whether a value makes an inequality true	**6.EE.B.5**
FORWARD 1	Solve multistep equations with the variable on both sides — near…	**8.EE.C.7.b**
FORWARD 2	Solve linear inequalities with the variable on both sides	**HSA-REI.B.3**

SET 34 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Sam has $40. He buys a $12 shirt and some pairs of socks that cost $3.50 each.

Write and solve an inequality for the number of pairs of socks p he can buy.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q1/5

1031

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

3.5p + 12 ≤ 40;
p ≤ 8

GRADING NOTE He can buy at most 8 pairs. Any correct inequality earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q1/5

Solve.

$-2x + 5 > 13$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q2/5

1032

QUESTION ID

S34-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$3x - 7 \geq 11$

A. $x \geq \frac{4}{3}$

B. $x \leq \frac{4}{3}$

C. $x \geq 6$

D. $x \leq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q3/5

1033

QUESTION ID

S34-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x ≥ 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q3/5

True or false?

Dividing both sides of $-4x < 20$ by -4 gives $x > -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q4/5

1034

QUESTION ID

S34-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q4/5

True or false?

The solution of $5x + 2 < 17$ is $x > 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q5/5

1035

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x < 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 1

Write an inequality for a situation • Grade 6 • Expressions & Equations • 6.EE.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Riders must be at least 48 inches tall.

Write an inequality for a rider's height h.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q1/5

1037

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

h ≥ 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an inequality for "a temperature t below −2°F."

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q2/5

1038

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

t < −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which inequality means "x is more than 7"?

A. $x < 7$

B. $x > 7$

C. $x \leq 7$

D. $x = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q3/5

1039

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x > 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q3/5

True or false?

$x < 3$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q4/5

1040

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q4/5

True or false?

5 is a solution of $x < 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q5/5

1041

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 2

Decide whether a value makes an inequality true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is 9 a solution of 2x > 16? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q1/5

1043

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(9) = 18, and 18 > 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q1/5

Which value makes y + 4 < 10 true?

A. 8

B. 10

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q2/5

1044

QUESTION ID

S34-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q2/5

Which of the values 3, 6, and 9 make $n - 2 \geq 4$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q3/5

1045

QUESTION ID

S34-B2-Q3

TYPE

Short answer

CORRECT ANSWER

6 and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q3/5

True or false?

x = 0 makes 3x < 1 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q4/5

1046

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q4/5

True or false?

k = 2 makes 5k > 10 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q5/5

1047

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b

FORWARD 1

Solve multistep equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Solve for x.

$0.5x + 7 = 2x - 2$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q1/5 • NEAREST RELATED

1049

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$\frac{2}{3}(x + 3) = x - 1$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q2/5 • NEAREST RELATED

1050

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q2/5 • NEAREST RELATED

Solve for x.

$6 - 3x = x + 22$

A. $x = -4$

B. $x = 7$

C. $x = -7$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q3/5 • NEAREST RELATED

1051

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$x = -2$ is the solution of $4(x + 5) = x + 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q4/5 • NEAREST RELATED

1052

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 1$ is the solution of $9x - 3 = 2x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q5/5 • NEAREST RELATED

1053

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 5/7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 34 • 7.EE.B.4.b

FORWARD 2

Solve linear inequalities with the variable on both sides • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve.

$5x - 3 > 2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q1/5

1055

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x > 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q1/5

Solve.

$-4(x + 1) \leq 2x + 8$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q2/5

1056

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \geq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q2/5

Solve.

$7 - 2x < x - 8$

A. $x < -5$

B. $x > -5$

C. $x > 5$

D. $x < 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q3/5

1057

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x > 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q3/5

True or false?

$x = 0$ is a solution of $3(x - 2) < x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q4/5

1058

QUESTION ID

S34-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

The solution of $6x + 1 \geq 4x + 11$ is $x \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q5/5

1059

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x \geq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Graph the solution set of an inequality and interpret it in context

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Graph an inequality on a number line	**6.EE.B.8**
BACKWARD 2	Read an inequality as positions on a number line	**6.NS.C.7.a**
FORWARD 1	Find the solution of a system of equations from its graph	**8.EE.C.8.a**
FORWARD 2	Graph the solutions of a linear inequality in two variables	**HSA-REI.D.12**

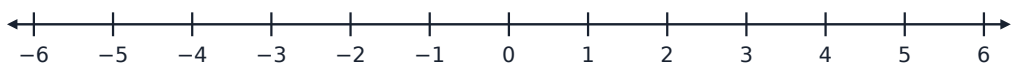
SET 35 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Graph the solution set of $x > -3$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q1/5

1062

QUESTION ID

S35-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

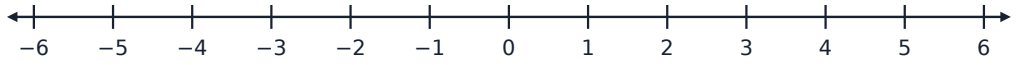
An open circle at −3 with an arrow or shading to the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 1 ≤ 9. Then graph the solution set on the number line.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q2/5

1063

QUESTION ID

S35-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A closed circle at 4 with an arrow or shading to the left

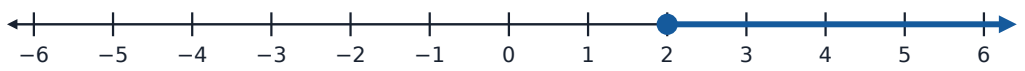
Written: x ≤ 4

GRADING NOTE Grade both: the solution x ≤ 4, and the graph with a closed circle at 4 shaded to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q2/5

Which inequality does the graph show?



A. $x > 2$

B. $x < 2$

C. $x \leq 2$

D. $x \geq 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q3/5

1064

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x \geq 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q3/5

A club has $30. It earns $5 for each ticket it sells. It needs more than $230.

Solve 5t + 30 > 230 and describe the solution in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q4/5

1065

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

t > 40. The club must sell more than 40 tickets (at least 41).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q4/5

True or false?

The graph shows the solution set of $-x < 2$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q5/5

1066

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

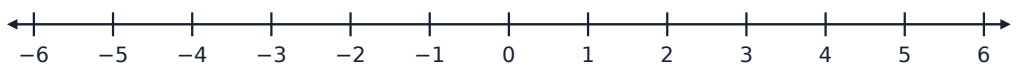
Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 1

Graph an inequality on a number line • Grade 6 • Expressions & Equations • 6.EE.B.8

Graph x < 4 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q1/5

1068

QUESTION ID

S35-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

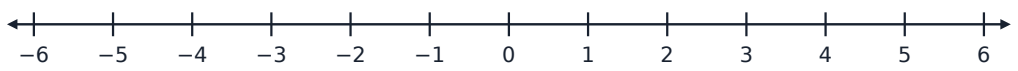
An open circle at 4 with an arrow or shading to the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q1/5

Graph x ≥ −1 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q2/5

1069

QUESTION ID

S35-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A closed circle at −1 with an arrow or shading to the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which inequality does the graph show?



A. $x < 0$

B. $x \geq 0$

C. $x \leq 0$

D. $x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q3/5

1070

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x < 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q3/5

True or false?

A closed circle on the graph of an inequality shows that the endpoint is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q4/5

1071

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The graph of $x > 5$ has a closed circle at 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q5/5

1072

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 2

Read an inequality as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

−4 is to the left of 1 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q1/5

1074

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−4 < 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q1/5

Which statement says that −2 is to the right of −6 on a number line?

A. −2 < −6

B. −2 = −6

C. −6 > −2

D. −2 > −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q2/5

1075

QUESTION ID

S35-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −2 > −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q2/5

What does $-8 < -3$ tell you about the positions of -8 and -3 on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q3/5

1076

QUESTION ID

S35-B2-Q3

TYPE

Short answer

CORRECT ANSWER

-8 is to the left of -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q3/5

True or false?

$-1 > -9$ because -1 is to the right of -9 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q4/5

1077

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q4/5

True or false?

$-5 > 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q5/5

1078

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b

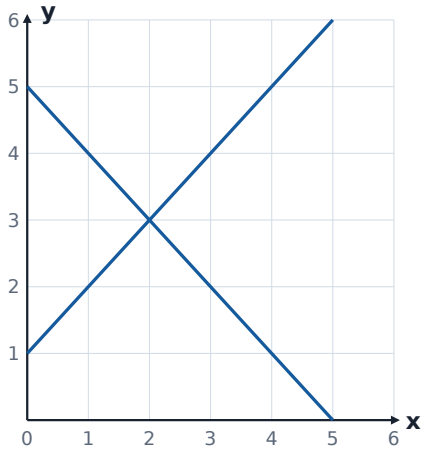
FORWARD 1

Find the solution of a system of equations from its graph • Grade 8 • Expressions & Equations • 8.EE.C.8.a

The graph shows y = x + 1 and y = −x + 5.

At what point do the lines intersect?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q1/5

1080

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q1/5

Two lines are graphed on the same grid.

What does their point of intersection represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q2/5

1081

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The pair (x, y) that makes both equations true; the solution of the system.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q2/5

The graphs of two linear equations are parallel lines.

How many solutions does the system have?

A. One solution

B. Two solutions

C. Infinitely many solutions

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q3/5

1082

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q3/5

True or false?

(1, 4) is the solution of the system y = 2x + 2 and y = 3x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q4/5

1083

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q4/5

True or false?

If two equations have the same graph, the system has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q5/5

1084

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q5/5

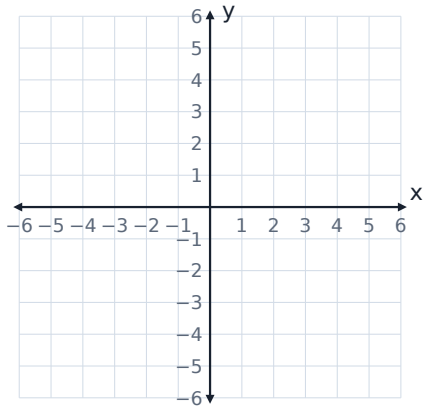
SET 35 • 7.EE.B.4.b

FORWARD 2

Graph the solutions of a linear inequality in two variables • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Graph y > x + 1 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q1/5

1086

QUESTION ID

S35-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A dashed line through (0, 1) with slope 1, with the region above the line shaded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q1/5

Is (2, 5) a solution of y ≥ 2x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q2/5

1087

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 5 ≥ 2(2), because 5 ≥ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q2/5

Which point is in the solution set of $y < -x + 3$?

A. (0, 0)

B. (0, 5)

C. (3, 1)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q3/5

1088

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q3/5

True or false?

The boundary line of the graph of $y \leq 3x - 2$ is drawn solid.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q4/5

1089

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q4/5

True or false?

The solution set of $y > 2$ is the region below the line $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q5/5

1090

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 7.G.A.1 • GEOMETRY

Find actual lengths and areas from a scale drawing

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Convert measurement units	**6.RP.A.3.d**
BACKWARD 2	Find the area of a rectangle	**4.MD.A.3**
FORWARD 1	Describe similar figures with dilations	**8.G.A.4**
FORWARD 2	Decide whether two triangles are similar	**HSG-SRT.A.2**

SET 36 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A scale drawing of a garden uses the scale 1 cm : 2 m. The garden is 6 cm long on the drawing.

What is the actual length of the garden?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q1/5

1093

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q1/5

A floor plan uses the scale 1 inch : 4 feet. A room is 3 inches by 2.5 inches on the plan.

What is the actual area of the room?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q2/5

1094

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

120 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q2/5

A map uses the scale 1 cm : 25 km. Two towns are 4.4 cm apart on the map.

What is the actual distance between the towns?

A. 100 km

B. 110 km

C. 5.68 km

D. 29.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q3/5

1095

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 110 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q3/5

True or false?

A drawing uses the scale 1 cm : 3 m. A 5 cm by 2 cm rectangle on the drawing has an actual area of 90 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q4/5

1096

QUESTION ID

S36-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q4/5

True or false?

A drawing uses the scale 1 cm : 10 cm. The actual area is 10 times the area on the drawing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q5/5

1097

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The actual area is 100 times the area on the drawing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q5/5

SET 36 • 7.G.A.1

BACKWARD 1

Convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

How many inches are in 4.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q1/5

1099

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q1/5

How many meters are in 350 centimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q2/5

1100

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q2/5

How many yards are in 27 feet?

A. 24

B. 9

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q3/5

1101

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q3/5

True or false?

2.5 kilometers = 2,500 meters

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q4/5

1102

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

6 feet = 60 inches

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q5/5

1103

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6 feet = 72 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 7.G.A.1

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 9 m wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q1/5

1105

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

108 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q1/5

A rectangle has an area of 84 square feet. It is 12 feet long.

How wide is it?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q2/5

1106

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q2/5

A rectangle is 15 cm long and 6 cm wide.

What is its area?

A. 96 square centimeters

B. 90 square centimeters

C. 21 square centimeters

D. 42 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q3/5

1107

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q3/5

True or false?

A square with 7-inch sides has an area of 49 square inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q4/5

1108

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A rectangle 10 m long and 4 m wide has an area of 28 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q5/5

1109

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 40 square meters; 28 m is its perimeter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 7.G.A.1

FORWARD 1

Describe similar figures with dilations • Grade 8 • Geometry • 8.G.A.4

A triangle has sides 3, 4, and 5 units. It is dilated by a scale factor of 2.5.

What are the new side lengths?

Side lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q1/5

1111

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.5, 10, and 12.5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q1/5

A 4-by-6 rectangle is similar to a rectangle that is 10 by w. The 4 side matches the 10 side.

Find w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q2/5

1112

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q2/5

A figure is dilated by a scale factor of $\frac{1}{3}$. One side is 18 cm long.

How long is the matching side of the image?

A. $18\frac{1}{3}$ cm

B. 3 cm

C. 6 cm

D. 54 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q3/5

1113

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q3/5

True or false?

Dilations keep angle measures the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q4/5

1114

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q4/5

True or false?

A 2-by-3 rectangle and a 4-by-5 rectangle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q5/5

1115

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 7.G.A.1

FORWARD 2

Decide whether two triangles are similar • Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has angles of 40°, 60°, and 80°. Triangle DEF has angles of 40°, 60°, and 80°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q1/5

1117

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Their corresponding angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q1/5

One triangle has sides 6, 8, and 10. Another has sides 9, 12, and 15.

What is the scale factor from the first triangle to the second?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q2/5

1118

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 3/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q2/5

Which two triangles must be similar?

A. Two right triangles

B. Two triangles that each have a 5 cm side

C. Two isosceles triangles

D. Two triangles that each have a 50° angle and a 70° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q3/5

1119

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Two triangles that each have a 50° angle and a 70° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q3/5

True or false?

Two triangles are similar with a scale factor of 3. Each pair of corresponding sides has a ratio of 3 : 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q4/5

1120

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q4/5

True or false?

All rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q5/5

1121

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q5/5

SET 37 • 7.G.A.1 • GEOMETRY

Reproduce a scale drawing at a different scale

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Draw a polygon on the coordinate plane	**6.G.A.3**
BACKWARD 2	Compare the size of a product to the size of one factor	**5.NF.B.5.a**
FORWARD 1	Describe the effect of a dilation on coordinates	**8.G.A.3**
FORWARD 2	Verify the properties of dilations	**HSG-SRT.A.1**

SET 37 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A drawing uses the scale 1 cm : 4 m. A wall is 5 cm long on the drawing.

How long should the wall be on a new drawing with the scale 1 cm : 2 m?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q1/5

1124

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

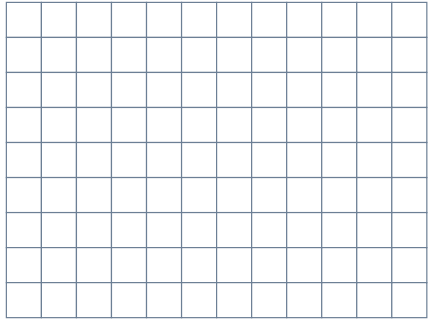
10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q1/5

A rectangle on a drawing is 4 units by 2 units.

Draw the rectangle at a scale twice as large on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q2/5

1125

QUESTION ID

S37-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A rectangle 8 units by 4 units, in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q2/5

A plan uses the scale 1 in : 6 ft. A table is 3 in long on the plan.

How long should the table be on a new plan with the scale 1 in : 3 ft?

A. 1.5 in

B. 18 in

C. 6 in

D. 9 in

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q3/5

1126

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6 in

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q3/5

True or false?

Changing a drawing's scale from 1 cm : 5 m to 1 cm : 10 m makes every length on the drawing half as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q4/5

1127

QUESTION ID

S37-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q4/5

True or false?

A 6 cm segment on a drawing with the scale 1 cm : 2 km becomes 9 cm on a drawing with the scale 1 cm : 3 km.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q5/5

1128

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It becomes 4 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q5/5

SET 37 • 7.G.A.1

BACKWARD 1

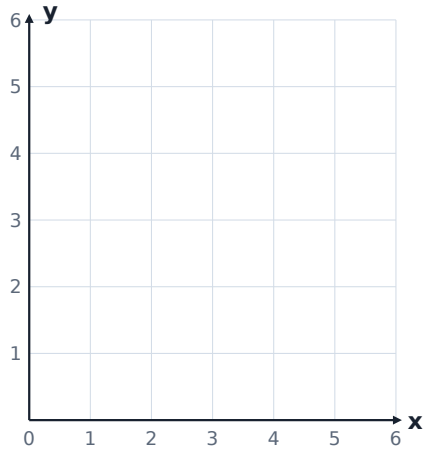
Draw a polygon on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Draw the polygon with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

1130

QUESTION ID

S37-B1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4)

Written: 4 units

GRADING NOTE Grade both: the rectangle drawn with the four given vertices, and the length 4 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

A rectangle has vertices (2, 1), (2, 6), (7, 6), and (7, 1).

How long is the side from (2, 1) to (7, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

1131

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

Which point makes a square with (1, 1), (4, 1), and (4, 4)?

A. (4, 7)

B. (3, 1)

C. (1, 3)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

1132

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

The points (0, 2) and (0, 7) are 5 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

1133

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

The side from (3, 2) to (3, 8) is 5 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

1134

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 6 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 7.G.A.1

BACKWARD 2

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

Without multiplying, is 3 × 12 greater than or less than 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q1/5

1136

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Greater than 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q1/5

Without multiplying, is $\frac{1}{2} \times 18$ greater than or less than 18?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q2/5

1137

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Less than 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q2/5

Which product is greater than 15?

A. 2×15

B. $\frac{3}{4} \times 15$

C. $\frac{1}{2} \times 15$

D. $\frac{1}{3} \times 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q3/5

1138

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 × 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} \times 9$ is less than 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q4/5

1139

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

4 × 8 is less than 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q5/5

1140

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 7.G.A.1

FORWARD 1

Describe the effect of a dilation on coordinates • Grade 8 • Geometry • 8.G.A.3

The point (4, −2) is dilated by a scale factor of 3 with the origin as the center.

What are the coordinates of the image?

Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q1/5

1142

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(12, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q1/5

A triangle has vertices (2, 2), (6, 2), and (2, 4). It is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the vertices of the image?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q2/5

1143

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 1), (3, 1), and (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q2/5

A dilation centered at the origin maps (5, 10) to (2, 4).

What is the scale factor?

A. $\frac{2}{5}$

B. $\frac{5}{2}$

C. -3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q3/5

1144

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A dilation by a scale factor of 2 with the origin as the center maps (−3, 1) to (−6, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q4/5

1145

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

A dilation by a scale factor of 4 with the origin as the center maps (2, 3) to (6, 7).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q5/5

1146

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It maps (2, 3) to (8, 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 7.G.A.1

FORWARD 2

Verify the properties of dilations • Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A segment 5 cm long is dilated by a scale factor of 1.8.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q1/5

1148

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q1/5

A line passes through the center of a dilation.

What is the image of the line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q2/5

1149

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The same line (it maps to itself).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q2/5

A line that does not pass through the center of a dilation is dilated.

What is its image?

A. The same line

B. A curve

C. A perpendicular line

D. A parallel line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q3/5

1150

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A parallel line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q3/5

True or false?

A dilation with a scale factor between 0 and 1 makes segments shorter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q4/5

1151

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q4/5

True or false?

A dilation always changes the slope of a line that does not pass through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q5/5

1152

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 7.G.A.2 • GEOMETRY

Draw triangles from given conditions

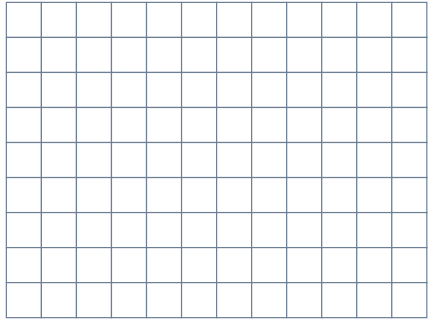
MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Use the Pythagorean theorem to find an unknown side length	**8.G.B.7**
FORWARD 2	Make formal constructions with a compass and straightedge	**HSG-CO.D.12**

SET 38 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

On the grid, draw a triangle with a right angle. The two sides that form the right angle should be 4 units and 6 units long.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q1/5

1155

QUESTION ID

S38-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Any right triangle whose two sides that form the right angle are 4 units and 6 units long, in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a ruler and a protractor. Draw a triangle with a 5 cm side that has a 50° angle at one end and a 60° angle at the other end.

Measure the third angle.

Third angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q2/5

1156

QUESTION ID

S38-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A triangle with a 5 cm side whose two end angles measure 50° and 60°, in any position or orientation

Written: 70°

GRADING NOTE Grade both: the triangle drawn with the given side and angles (within ±2 mm and ±2°), and the third angle 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q2/5

Which three side lengths can form a triangle?

A. 1 cm, 2 cm, 3 cm

B. 2 cm, 3 cm, 6 cm

C. 5 cm, 7 cm, 10 cm

D. 4 cm, 4 cm, 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q3/5

1157

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

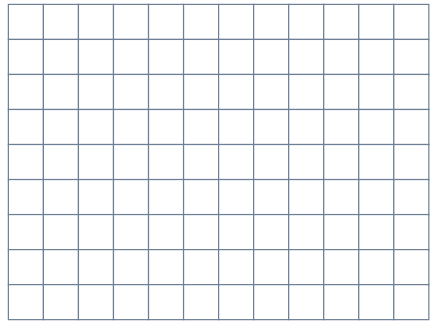
C. 5 cm, 7 cm, 10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q3/5

On the grid, draw an isosceles triangle that is not equilateral.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q4/5

1158

QUESTION ID

S38-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Any triangle with exactly two sides of equal length (the third side has a different length)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q4/5

True or false?

A triangle can have angles of 90°, 60°, and 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q5/5

1159

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The angles of a triangle add to 180°, and these add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q5/5

SET 38 • 7.G.A.2

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

Use a protractor to draw a 65° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q1/5

1161

QUESTION ID

S38-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

An angle that measures 65° (within ±2°), in any orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q1/5

One ray of an angle lines up with 0 on a protractor. The other ray passes through 120.

What is the measure of the angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q2/5

1162

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q2/5

Which angle is greater than a right angle?

A. 80°

B. 90°

C. 110°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q3/5

1163

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q3/5

True or false?

A protractor measures angles in degrees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q4/5

1164

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A 30° angle is an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q5/5

1165

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 7.G.A.2

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has a 90° angle.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q1/5

1167

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q1/5

A triangle has angles of 100°, 50°, and 30°.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q2/5

1168

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Obtuse triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q2/5

What kind of triangle has three angles that are each less than 90°?

A. Obtuse triangle

B. Acute triangle

C. Straight triangle

D. Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q3/5

1169

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Acute triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q3/5

True or false?

An obtuse triangle has exactly one obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q4/5

1170

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q5/5

1171

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 7.G.A.2

FORWARD 1

Use the Pythagorean theorem to find an unknown side length • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.

How long is the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q1/5

1173

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q1/5

A right triangle has a hypotenuse of 13 m and one leg of 5 m.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q2/5

1174

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q2/5

A 10-foot ladder leans against a wall. The top of the ladder is 8 feet above the ground.

How far is the bottom of the ladder from the wall?

A. 6 feet

B. 12.8 feet

C. 18 feet

D. 2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q3/5

1175

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q3/5

True or false?

A right triangle with legs of 9 and 12 has a hypotenuse of 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q4/5

1176

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs of 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q5/5

1177

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hypotenuse is √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 7.G.A.2

FORWARD 2

Make formal constructions with a compass and straightedge • Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12

To construct an equilateral triangle on segment AB, you draw two circles.

What are the centers and the radius of the circles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q1/5

1179

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The centers are A and B, and each radius is AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q1/5

To construct the perpendicular bisector of a segment, where do you place the compass point for the two arcs?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q2/5

1180

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At each endpoint of the segment, with the same radius (more than half the segment).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q2/5

1180

Which tool draws arcs and circles in a formal construction?

A. Protractor

B. Compass

C. Ruler with marks

D. Calculator

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q3/5

1181

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Compass

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q3/5

True or false?

The perpendicular bisector of a segment passes through its midpoint and makes right angles with it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q4/5

1182

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q4/5

True or false?

A formal construction may use a protractor to measure angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q5/5

1183

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q5/5

SET 39 • 7.G.A.2 • GEOMETRY

Decide whether given conditions make one triangle, more than one triangle, or no triangle

MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Find an unknown angle when angles are joined	**4.MD.C.7**
BACKWARD 2	Find the perimeter of a polygon from its side lengths	**3.MD.D.8**
FORWARD 1	Use the converse of the Pythagorean theorem to test for a right tr…	**8.G.B.6**
FORWARD 2	Use triangle congruence criteria	**HSG-CO.B.8**

SET 39 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

How many triangles have sides of 3 cm, 4 cm, and 9 cm?

A. Exactly one triangle

B. More than one triangle

C. No triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q1/5

1186

QUESTION ID

S39-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. No triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q1/5

A triangle has angles of 30°, 60°, and 90°. No side lengths are given.

How many different triangles fit these conditions? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q2/5

1187

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

More than one. Triangles of different sizes can have the same three angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q2/5

How many different triangles have sides of 5 cm, 6 cm, and 7 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q3/5

1188

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

Exactly one triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q3/5

True or false?

Angles of 80° and 110° cannot both be angles of the same triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q4/5

1189

QUESTION ID

S39-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q4/5

True or false?

Three side lengths that form a triangle can make two triangles with different shapes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q5/5

1190

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q5/5

SET 39 • 7.G.A.2

BACKWARD 1

Find an unknown angle when angles are joined • Grade 4 • Measurement & Data • 4.MD.C.7

Angle ABC is split into two angles that measure 35° and 50°.

What is the measure of angle ABC?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q1/5

1192

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q1/5

A 90° angle is split into two angles. One measures 28°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q2/5

1193

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

62°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q2/5

A 120° angle is split into two angles. One measures 45°.

What does the other angle measure?

A. 45°

B. 165°

C. 75°

D. 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q3/5

1194

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 75°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A 45° angle and a 45° angle joined together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q4/5

1195

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A 120° angle and a 70° angle joined together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q5/5

1196

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make a 190° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 7.G.A.2

BACKWARD 2

Find the perimeter of a polygon from its side lengths • Grade 3 • Measurement & Data • 3.MD.D.8

A triangle has sides of 7 cm, 9 cm, and 12 cm.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q1/5

1198

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

28 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q1/5

A triangle has a perimeter of 30 inches. Two of its sides are 8 inches and 10 inches.

How long is the third side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q2/5

1199

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q2/5

A triangle has sides of 5 m, 5 m, and 6 m.

What is its perimeter?

A. 11 m

B. 21 m

C. 150 m

D. 16 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q3/5

1200

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 16 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A triangle with sides of 4 feet, 6 feet, and 7 feet has a perimeter of 17 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q4/5

1201

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A triangle with three 9 cm sides has a perimeter of 18 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q5/5

1202

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 7.G.A.2

FORWARD 1

Use the converse of the Pythagorean theorem to test for a right triangle • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides of 7, 24, and 25 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q1/5

1204

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

$7^2 + 24^2 = 49 + 576 = 625 = 25^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q1/5

Is a triangle with sides of 6, 7, and 9 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q2/5

1205

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. $6^2 + 7^2 = 85$, but $9^2 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q2/5

Which side lengths form a right triangle?

A. 8, 15, 17

B. 4, 5, 6

C. 5, 6, 8

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q3/5

1206

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8, 15, 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q3/5

True or false?

A triangle with sides of 9, 40, and 41 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q4/5

1207

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q4/5

True or false?

A triangle with sides of 3, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q5/5

1208

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 7.G.A.2

FORWARD 2

Use triangle congruence criteria • Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8

Two triangles have three pairs of equal sides.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q1/5

1210

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q1/5

Two triangles have two pairs of equal sides and equal angles between those sides.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q2/5

1211

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q2/5

Which information is NOT enough to show that two triangles are congruent?

A. Two angles and the side between them (ASA)

B. Two sides and the angle between them (SAS)

C. Three pairs of equal sides (SSS)

D. Three pairs of equal angles (AAA)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q3/5

1212

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Three pairs of equal angles (AAA)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q3/5

True or false?

SSS shows that three side lengths determine one triangle, up to congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q4/5

1213

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of equal sides are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q5/5

1214

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 7.G.A.3 • GEOMETRY

Describe the cross sections made by slicing right prisms and pyramids

MAIN	Grade 7 main question	**7.G.A.3**
BACKWARD 1	Identify the faces of solids from their nets	**6.G.A.4**
BACKWARD 2	Name shapes by their number of sides and faces	**2.G.A.1**
FORWARD 1	Find the volumes of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Identify cross sections of three-dimensional objects	**HSG-GMD.B.4**

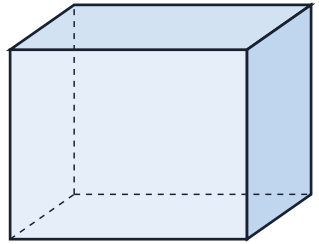
SET 40 • 7.G.A.3

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane parallel to its base.

What is the shape of the cross section?



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q1/5

1217

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q1/5

A right square pyramid is sliced by a plane parallel to its base.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q2/5

1218

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

A square smaller than the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q2/5

A cube is sliced by a plane that is parallel to one of its faces.

What is the shape of the cross section?

A. A triangle

B. A pentagon

C. A square

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q3/5

1219

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q3/5

True or false?

A right square pyramid is sliced by a plane through its top vertex that is perpendicular to the base. The cross section is a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q4/5

1220

QUESTION ID

S40-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q4/5

True or false?

A right rectangular prism sliced by a plane parallel to its base can have a triangle as its cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q5/5

1221

QUESTION ID

S40-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q5/5

SET 40 • 7.G.A.3

BACKWARD 1

Identify the faces of solids from their nets • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q1/5

A net has 1 square and 4 triangles.

What solid does it fold into?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A square pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q2/5

Which solid has a net of 2 triangles and 3 rectangles?

A. Square pyramid

B. Rectangular prism

C. Triangular prism

D. Cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Triangular prism

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

The net of a cube has 6 squares.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A triangular pyramid has 5 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 4 faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 7.G.A.3

BACKWARD 2

Name shapes by their number of sides and faces • Grade 2 • Geometry • 2.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

How many sides does a pentagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q1/5

A shape has 6 sides and 6 angles.

What is the name of the shape?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q2/5

Which shape has 4 sides?

A. Quadrilateral

B. Hexagon

C. Pentagon

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Quadrilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q3/5

True or false?

A cube has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q4/5

True or false?

A triangle has 4 angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 7.G.A.3

FORWARD 1

Find the volumes of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 2 cm and a height of 9 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q1/5

1235

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

113.04 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q1/5

A cone has a radius of 6 inches and a height of 5 inches.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q2/5

1236

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q2/5

A sphere has a radius of 3 m.

What is its volume in terms of π?

A. 12π cubic meters

B. 108π cubic meters

C. 27π cubic meters

D. 36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q3/5

1237

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q3/5

True or false?

A cone and a cylinder have the same base and height. The cone's volume is $\frac{1}{3}$ of the cylinder's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q4/5

1238

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q5/5

1239

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 20π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 7.G.A.3

FORWARD 2

Identify cross sections of three-dimensional objects • Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A cylinder is sliced by a plane parallel to its base.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q1/5

1241

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q1/5

A cone is sliced by a plane through its top point (apex) that is perpendicular to its base.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q2/5

1242

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle (an isosceles triangle)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q2/5

A sphere is sliced by a plane.

What is the shape of the cross section?

A. A triangle

B. A square

C. An oval that is not a circle

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q3/5

1243

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q3/5

True or false?

Spinning a rectangle around one of its sides makes a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q4/5

1244

QUESTION ID

S40-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q4/5

True or false?

A cube can never have a cross section shaped like a hexagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q5/5

1245

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A slice through the midpoints of six edges makes a hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 7.G.B.4 • GEOMETRY

Find the circumference of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the perimeter of a rectangle	**4.MD.A.3**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Find the volume of a cylinder	**8.G.C.9**
FORWARD 2	Find the length of an arc of a circle	**HSG-C.B.5**

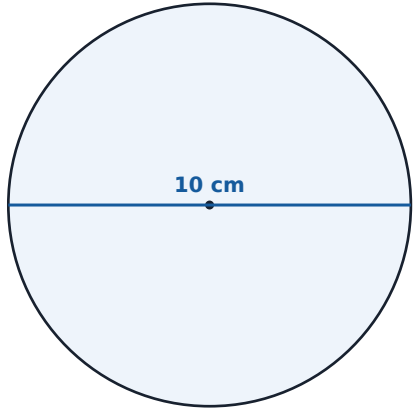
SET 41 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the circumference of the circle. Use 3.14 for π.

Circumference: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q1/5

1248

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

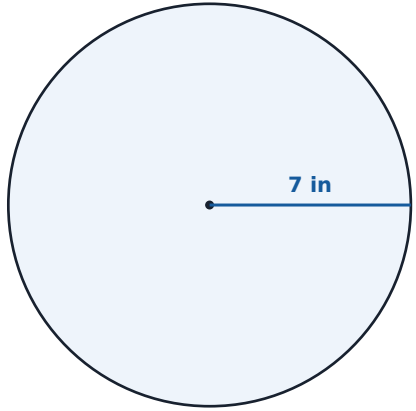
31.4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q1/5

Find the circumference of the circle. Use $\frac{22}{7}$ for π.



Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q2/5

1249

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

44 in

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q2/5

A circular track has a circumference of 62.8 m.

What is its diameter? Use 3.14 for π.

A. 31.4 m

B. 20 m

C. 10 m

D. 197.2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q3/5

1250

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q3/5

True or false?

A wheel has a radius of 15 inches. In one full turn it rolls about 94.2 inches. (Use 3.14 for π.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q4/5

1251

QUESTION ID

S41-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q4/5

True or false?

A circle with a radius of 4 m has a circumference of 4π m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q5/5

1252

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The circumference is 8π m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q5/5

SET 41 • 7.G.B.4

BACKWARD 1

Find the perimeter of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle is 9 m long and 4 m wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q1/5

1254

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

26 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q1/5

A square has 8 cm sides.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q2/5

1255

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q2/5

A rectangle is 12 feet long and 5 feet wide.

What is its perimeter?

A. 29 feet

B. 17 feet

C. 60 feet

D. 34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q3/5

1256

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q3/5

True or false?

A rectangle 6 inches long and 3 inches wide has a perimeter of 18 inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q4/5

1257

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q4/5

True or false?

A square with 10 m sides has a perimeter of 100 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q5/5

1258

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter is 40 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 7.G.B.4

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

3.14 × 6

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q1/5

1260

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q1/5

Multiply.

3.14 × 2.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q2/5

1261

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7.85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

3.14×12

A. 37.68

B. 36.48

C. 376.8

D. 15.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q3/5

1262

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 37.68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

$3.14 \times 10 = 31.4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q4/5

1263

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

$3.14 \times 4 = 12.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q5/5

1264

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 7.G.B.4

FORWARD 1

Find the volume of a cylinder • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 5 feet and a height of 4 feet.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q1/5

1266

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

100π cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q1/5

A cylinder has a diameter of 6 cm and a height of 10 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q2/5

1267

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

282.6 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q2/5

A cylinder has a volume of 90π cubic meters and a radius of 3 m.

What is its height?

A. 30 m

B. 3 m

C. 10 m

D. 15 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q3/5

1268

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q3/5

True or false?

Doubling the height of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q4/5

1269

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 1 and a height of 1 has a volume of 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q5/5

1270

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 7.G.B.4

FORWARD 2

Find the length of an arc of a circle • Grade 9 (High School) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 6 cm.

Find the length of an arc with a central angle of 90°, in terms of π.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q1/5

1272

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3π cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q1/5

A circle has a circumference of 40 inches.

How long is an arc with a central angle of 45°?

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q2/5

1273

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q2/5

A circle has a radius of 10 m.

What is the length of an arc with a central angle of 60°?

A. $\frac{20}{3}\pi$ m

B. $\frac{5}{3}\pi$ m

C. $\frac{10}{3}\pi$ m

D. 60π m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q3/5

1274

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10/3π m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q3/5

True or false?

An arc with a central angle of 180° is half of the circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q4/5

1275

QUESTION ID

S41-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 3, an arc with a central angle of 120° has a length of 3π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q5/5

1276

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The arc length is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q5/5

SET 42 • 7.G.B.4 • GEOMETRY

Find the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Evaluate powers with whole-number exponents	**6.EE.A.1**
BACKWARD 2	Evaluate a formula	**6.EE.A.2.c**
FORWARD 1	Find the volume of a cone	**8.G.C.9**
FORWARD 2	Find the area of a sector of a circle	**HSG-C.B.5**

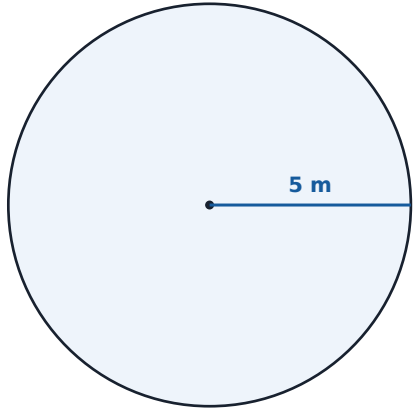
SET 42 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the area of the circle.
Use 3.14 for π.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q1/5

1279

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

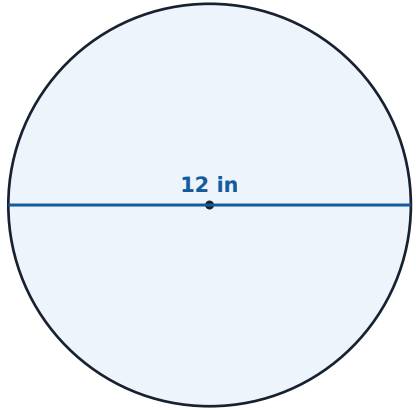
78.5 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q1/5

Find the area of the circle in terms of π.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q2/5

1280

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

36π square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q2/5

A circular rug has a radius of 3 feet.

What is its area? Use 3.14 for π.

A. 113.04 square feet

B. 28.26 square feet

C. 9.42 square feet

D. 18.84 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q3/5

1281

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 28.26 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q3/5

True or false?

A circle with a radius of 10 cm has an area of 100π square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q4/5

1282

QUESTION ID

S42-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q4/5

True or false?

A circle with a diameter of 8 m has an area of 64π square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q5/5

1283

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The radius is 4 m, so the area is 16π square meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q5/5

SET 42 • 7.G.B.4

BACKWARD 1

Evaluate powers with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q1/5

1285

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q2/5

1286

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q2/5

Find the value.

4^3

A. 7

B. 64

C. 12

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q3/5

1287

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q4/5

1288

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q5/5

1289

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 7.G.B.4

BACKWARD 2

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Use $A = s^2$.

Find A when s = 11.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q1/5

1291

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q1/5

Use $A = 3.14r^2$.

Find A when r = 2.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q2/5

1292

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12.56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $A = \frac{1}{2}bh$.

Find A when b = 8 and h = 5.

A. 13

B. 20

C. 40

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q3/5

1293

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q3/5

True or false?

Use $V = s^3$. When $s = 2$, $V = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q4/5

1294

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Use A = lw. When l = 3.5 and w = 2, A = 5.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q5/5

1295

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 7.G.B.4

FORWARD 1

Find the volume of a cone • Grade 8 • Geometry • 8.G.C.9

A cone has a radius of 3 cm and a height of 8 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q1/5

1297

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

24π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q1/5

A cone has a radius of 6 feet and a height of 7 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q2/5

1298

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

263.76 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q2/5

A cone has a base area of 30 square inches and a height of 9 inches.

What is its volume?

A. 270 cubic inches

B. 90 cubic inches

C. 30 cubic inches

D. 39 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q3/5

1299

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

A cone with a radius of 3 and a height of 3 has a volume of 9π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q4/5

1300

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q4/5

True or false?

The volume of a cone is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q5/5

1301

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The volume of a cone is (1/3)$\pi r^2 h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 7.G.B.4

FORWARD 2

Find the area of a sector of a circle • Grade 9 (High School) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 4 m.

Find the area of a sector with a central angle of 90°, in terms of π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q1/5

1303

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q1/5

A pizza with a radius of 8 inches is cut into 8 equal slices.

What is the area of one slice, in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q2/5

1304

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8π square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q2/5

A circle has a radius of 6 cm.

What is the area of a sector with a central angle of 120°?

A. 12π square centimeters **B.** 24π square centimeters

C. 4π square centimeters **D.** 36π square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q3/5

1305

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12π square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q3/5

True or false?

A sector with a central angle of 180° has half the area of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q4/5

1306

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 6, a sector with a central angle of 60° has an area of 12π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q5/5

1307

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area is 6π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 43 • 7.G.B.4 • GEOMETRY

Explain the relationship between the circumference and the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the area of a parallelogram	**6.G.A.1**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Use square roots to solve equations of the form $x^2 = p$	**8.EE.A.2**
FORWARD 2	Give an informal argument for circle and volume formulas	**HSG-GMD.A.1**

SET 43 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle with radius r is cut into many equal wedges. The wedges are rearranged into a shape close to a parallelogram. Its base is half the circumference.

Write the base in terms of r.

Base: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q1/5

1310

QUESTION ID
S43-M-Q1

TYPE
Short answer

CORRECT ANSWER
πr

GRADING NOTE Half of 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q1/5

A circle with radius r is cut into many equal wedges and rearranged into a shape close to a parallelogram.

What is the height of the parallelogram in terms of r?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q2/5

1311

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q2/5

A circle rearranged into a near-parallelogram has a base of πr and a height of r.

What is the area?

A. πr^2

B. $2\pi r^2$

C. πr

D. $2\pi r$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q3/5

1312

QUESTION ID

S43-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. πr^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q3/5

True or false?

A circle has circumference C and radius r. Its area equals $\frac{1}{2} \times C \times r$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q4/5

1313

QUESTION ID

S43-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q4/5

True or false?

A circle with a circumference of 10π has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q5/5

1314

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its radius is 5, so its area is 25π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q5/5

SET 43 • 7.G.B.4

BACKWARD 1

Find the area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

A parallelogram has a base of 9 cm and a height of 4 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A parallelogram has a base of 12 m and a height of 6.5 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

78 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A triangle is cut off one end of a parallelogram and moved to the other end to make a new shape.

What shape does it make?

A. A pentagon

B. A trapezoid that is not a rectangle

C. A circle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

A parallelogram and a rectangle with the same base and the same height have the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A parallelogram has a base of 10 and slanted sides of 6. Its area is 60.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area uses the height, not the slanted side, so it is less than 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 7.G.B.4

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Is $\frac{1}{2} \times 2 \times r$ equivalent to r? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 1/2 × 2 = 1, and 1 × r = r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q1/5

Which expression is equivalent to 3 × r × r?

A. 6r

B. $3 + r^2$

C. $(3r)^2$

D. $3r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S43-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $3r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression equivalent to $x \cdot x \cdot x$ using an exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S43-B2-Q3

TYPE

Short answer

CORRECT ANSWER

x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q3/5

True or false?

2(4y) and 8y are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q4/5

True or false?

a^2 and $2a$ are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 7.G.B.4

FORWARD 1

Use square roots to solve equations of the form $x^2 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x.

$x^2 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q1/5

1328

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 or −8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q1/5

A circle has an area of 49π square centimeters.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q2/5

1329

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q2/5

Solve for x.

$x^3 = 27$

A. x = 3

B. x = 9

C. x = 6

D. x = 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q3/5

1330

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{144} = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q4/5

1331

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q4/5

True or false?

√50 is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q5/5

1332

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 7.G.B.4

FORWARD 2

Give an informal argument for circle and volume formulas • Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Use the wedge argument to explain why the area of a circle is πr^2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q1/5

1334

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The wedges form a shape close to a parallelogram with base πr (half the circumference) and height r, so the area is $\pi r \times r = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q1/5

Two stacks of 20 identical coins have the same height. One stack is straight and one is slanted.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q2/5

1335

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. At every height the cross sections have the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q2/5

The circumference of a circle is $C = \pi d$.

What is the ratio of C to d?

A. 2π

B. π

C. 2

D. r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q3/5

1336

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q3/5

True or false?

As a circle is cut into more and more wedges, the rearranged shape gets closer to a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q4/5

1337

QUESTION ID

S43-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q4/5

True or false?

Cavalieri's principle says that any two solids with the same height have the same volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q5/5

1338

QUESTION ID
S43-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The cross sections must also have equal areas at every height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q5/5

SET 44 • 7.G.B.5 • GEOMETRY

Write and solve equations for supplementary and complementary angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Find an unknown part of a right angle or a straight angle	**4.MD.C.7**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Use the angle sum and exterior angles of triangles	**8.G.A.5**
FORWARD 2	Prove theorems about angles formed by parallel lines	**HSG-CO.C.9**

SET 44 • 7.G.B.5

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

Two angles are complementary. One measures 37°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q1/5

1341

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

53°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q1/5

Angles that measure (2x + 10)° and 70° are supplementary.

Write and solve an equation for x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q2/5

1342

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

2x + 10 + 70 = 180; x = 50

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q2/5

Two complementary angles measure x° and (x + 20)°.

What is the value of x?

A. 45

B. 35

C. 80

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q3/5

1343

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q3/5

True or false?

An angle of 125° and an angle of 55° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q4/5

1344

QUESTION ID

S44-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Angles of 45° and 55° are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q5/5

1345

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 100°, not 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q5/5

SET 44 • 7.G.B.5

BACKWARD 1

Find an unknown part of a right angle or a straight angle • Grade 4 • Measurement & Data • 4.MD.C.7

A straight angle (180°) is split into two angles. One measures 115°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q1/5

1347

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q1/5

A right angle is split into two angles. One measures 34°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q2/5

1348

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

56°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q2/5

A right angle is split into three angles. Two of them measure 20° and 30°.

What does the third angle measure?

A. 50°

B. 60°

C. 40°

D. 10°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q3/5

1349

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q3/5

True or false?

Angles of 72° and 18° together make a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q4/5

1350

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Angles of 95° and 95° together make a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q5/5

1351

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 7.G.B.5

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 48 = 90

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q1/5

1353

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q1/5

Solve for x.

3x = 180

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q2/5

1354

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for y.

125 + y = 180

A. y = 55

B. y = 45

C. y = 65

D. y = 305

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q3/5

1355

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q3/5

True or false?

a = 27 is the solution of a + 63 = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q4/5

1356

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q4/5

True or false?

b = 80 is the solution of 2b = 180.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q5/5

1357

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 7.G.B.5

FORWARD 1

Use the angle sum and exterior angles of triangles • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle measure 48° and 75°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q1/5

1359

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

57°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 130°. One of the two remote interior angles measures 60°.

What does the other remote interior angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q2/5

1360

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q2/5

The angles of a triangle measure x°, 2x°, and 3x°.

What is the value of x?

A. 36

B. 30

C. 45

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q3/5

1361

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The interior angles of a triangle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q4/5

1362

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q5/5

1363

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 7.G.B.5

FORWARD 2

Prove theorems about angles formed by parallel lines • Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9

Two parallel lines are cut by a transversal. One angle measures 65°.

What does its alternate interior angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q1/5

1365

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q1/5

Two parallel lines are cut by a transversal. Two same-side interior angles measure 3x° and 2x°.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q2/5

1366

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q2/5

Which statement is a theorem?

A. Vertical angles are complementary.

B. Vertical angles are supplementary.

C. Vertical angles are congruent.

D. Vertical angles are always right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q3/5

1367

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Vertical angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q3/5

True or false?

When parallel lines are cut by a transversal, corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q4/5

1368

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

Any two lines cut by a transversal form congruent alternate interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q5/5

1369

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE This is true only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 7.G.B.5 • GEOMETRY

Use vertical and adjacent angles to find unknown angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Identify lines, rays, segments, and angles	**4.G.A.1**
BACKWARD 2	Understand angle measure as a fraction of a full turn	**4.MD.C.5.a**
FORWARD 1	Find angles formed by parallel lines cut by a transversal	**8.G.A.5**
FORWARD 2	Prove that vertical angles are congruent	**HSG-CO.C.9**

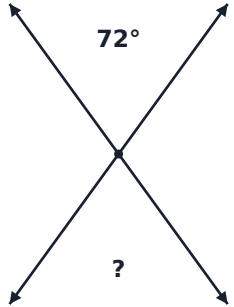
SET 45 • 7.G.B.5

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. One angle measures 72°.

What does the angle marked with a question mark measure?



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q1/5

1372

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

72°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q1/5

Two lines intersect. One angle measures (3x + 15)°. The angle vertical to it measures 75°.

What is the value of x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q2/5

1373

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q2/5

Two lines intersect. Angles a and b are adjacent and form a straight line. Angle a measures 118°.

What does angle b measure?

A. 72°

B. 118°

C. 62°

D. 242°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q3/5

1374

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 62°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q3/5

True or false?

Adjacent angles share a vertex and a side and do not overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q4/5

1375

QUESTION ID

S45-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q4/5

True or false?

Vertical angles are always adjacent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q5/5

1376

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q5/5

SET 45 • 7.G.B.5

BACKWARD 1

Identify lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

How many endpoints does a ray have?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q1/5

1378

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q1/5

What figure is formed by two rays that share an endpoint?

Figure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q2/5

1379

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

An angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q2/5

Which figure goes on forever in both directions?

A. Ray

B. Segment

C. Line

D. Angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q3/5

1380

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q3/5

True or false?

Two lines that cross meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q4/5

1381

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

A line segment has no endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q5/5

1382

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 7.G.B.5

BACKWARD 2

Understand angle measure as a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many degrees are in one full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q1/5

1384

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An angle turns through $\frac{1}{4}$ of a circle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q2/5

1385

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q2/5

An angle turns through $\frac{1}{2}$ of a circle.

What is its measure?

A. 90°

B. 45°

C. 360°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q3/5

1386

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q3/5

True or false?

An angle that turns through $\frac{1}{360}$ of a circle measures 1°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q4/5

1387

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q4/5

True or false?

An angle that turns through $\frac{1}{3}$ of a circle measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q5/5

1388

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It measures 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 7.G.B.5

FORWARD 1

Find angles formed by parallel lines cut by a transversal • Grade 8 • Geometry • 8.G.A.5

Two parallel lines are cut by a transversal. One angle measures 110°.

What does the corresponding angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q1/5

1390

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. Alternate interior angles measure (4x − 10)° and 70°.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q2/5

1391

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q2/5

Two parallel lines are cut by a transversal. One same-side interior angle measures 75°.

What does the other measure?

A. 15°

B. 105°

C. 75°

D. 285°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q3/5

1392

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q3/5

True or false?

When parallel lines are cut by a transversal, alternate exterior angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q4/5

1393

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q4/5

True or false?

When parallel lines are cut by a transversal, corresponding angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q5/5

1394

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 7.G.B.5

FORWARD 2

Prove that vertical angles are congruent • Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 form a linear pair. Angles 2 and 3 form a linear pair.

Explain why angle 1 and angle 3 have the same measure.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q1/5

1396

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each one equals 180° minus the measure of angle 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q1/5

Two lines intersect. Vertical angles measure $(5x - 20)°$ and $(3x + 30)°$.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q2/5

1397

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q2/5

Angles 1 and 2 form a linear pair.

Which reason supports angle 1 + angle 2 = 180°?

A. The angles of a triangle add to 180°.

B. Vertical angles are congruent.

C. Corresponding angles are congruent.

D. Linear pairs are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q3/5

1398

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Linear pairs are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q3/5

True or false?

Two angles that are each supplementary to the same angle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q4/5

1399

QUESTION ID

S45-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q4/5

True or false?

Two intersecting lines always form vertical angles that add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q5/5

1400

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 7.G.B.6 • GEOMETRY

Find the area of composite figures made of triangles and quadrilaterals

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the area of a triangle	**6.G.A.1**
BACKWARD 2	Find areas of rectilinear figures by breaking them into rectangles	**3.MD.C.7.d**
FORWARD 1	Find the volume of a solid made of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Use geometric shapes to model real objects	**HSG-MG.A.1**

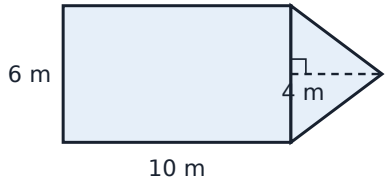
SET 46 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

The figure is a 10 m by 6 m rectangle and a triangle with a base of 6 m (the rectangle's side) and a height of 4 m. The parts do not overlap.

What is the total area?



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q1/5

1403

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

72 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q1/5

An L-shaped floor is made of two rectangles that do not overlap. One is 8 feet by 5 feet and the other is 4 feet by 3 feet.

What is the area of the floor?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q2/5

1404

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

52 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q2/5

A trapezoid has parallel sides of 8 cm and 12 cm. The distance between them (the height) is 5 cm.

What is its area?

A. 100 square centimeters

B. 480 square centimeters

C. 50 square centimeters

D. 60 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q3/5

1405

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 50 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q3/5

True or false?

A 4 cm by 4 cm square is cut from one corner of a 12 cm by 8 cm rectangle. The area that is left is 80 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q4/5

1406

QUESTION ID

S46-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q4/5

True or false?

A 9 m by 4 m rectangle and a right triangle with legs of 4 m and 3 m are joined without overlapping. Their total area is 48 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q5/5

1407

QUESTION ID

S46-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total area is 36 + 6 = 42 square meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q5/5

SET 46 • 7.G.B.6

BACKWARD 1

Find the area of a triangle • Grade 6 • Geometry • 6.G.A.1

A triangle has a base of 14 cm and a height of 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q1/5

1409

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

35 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q1/5

A right triangle has legs of 9 m and 6 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q2/5

1410

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q2/5

A triangle has a base of 7 feet and a height of 10 feet.

What is its area?

A. 17 square feet

B. 35 square feet

C. 140 square feet

D. 70 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q3/5

1411

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A triangle with a base of 8 and a height of 3 has an area of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q4/5

1412

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

A triangle with a base of 6 and a height of 6 has an area of 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q5/5

1413

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 7.G.B.6

BACKWARD 2

Find areas of rectilinear figures by breaking them into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

A shape is made of a 5-by-3 rectangle and a 2-by-2 square that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q1/5

1415

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

19 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q1/5

A shape is made of a 6-by-2 rectangle and a 3-by-4 rectangle that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q2/5

1416

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q2/5

A shape is made of a 4-by-4 square and a 4-by-1 rectangle that do not overlap.

What is its area?

A. 16 square units

B. 20 square units

C. 13 square units

D. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q3/5

1417

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q3/5

True or false?

A 3-by-5 rectangle and a 2-by-5 rectangle that do not overlap have a total area of 25 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q4/5

1418

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q4/5

True or false?

A 7-by-2 rectangle and a 1-by-3 rectangle that do not overlap have a total area of 13 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q5/5

1419

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total area is 17 square units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 7.G.B.6

FORWARD 1

Find the volume of a solid made of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

ANSWER KEY Reduced copy of the question page, with its answer

A cone sits on top of a cylinder. Both have a radius of 3. The cylinder is 10 tall and the cone is 4 tall.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q1/5

1421

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

102π

GRADING NOTE 90π + 12π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q1/5

A hemisphere (half of a sphere) has a radius of 6 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q2/5

1422

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

144π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q2/5

A capsule is a cylinder with radius 2 and height 5 with half of a sphere of radius 2 on each end.

What is its volume?

A. $\frac{52}{3}\pi$

B. 20π

C. $\frac{92}{3}\pi$

D. $\frac{32}{3}\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q3/5

1423

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 92/3π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q3/5

True or false?

A sphere with a radius of 3 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q4/5

1424

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q4/5

True or false?

A cone with a radius of 3 and a height of 4 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q5/5

1425

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 12π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 7.G.B.6

FORWARD 2

Use geometric shapes to model real objects • Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1

A tree trunk is modeled as a cylinder with a diameter of 2 feet and a height of 30 feet.

Estimate its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q1/5

1427

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

30π cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q1/5

A round tabletop is modeled as a circle with a circumference of 9.42 feet.

What is its radius? Use 3.14 for π.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q2/5

1428

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q2/5

Which shape is the best model for a basketball?

A. Cylinder

B. Sphere

C. Cube

D. Cone

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q3/5

1429

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q3/5

True or false?

A soup can is well modeled by a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q4/5

1430

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q4/5

True or false?

An ice cream cone is best modeled by a rectangular prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q5/5

1431

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 7.G.B.6 • GEOMETRY

Find the volume of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the volume of a rectangular prism with fractional edges	**6.G.A.2**
BACKWARD 2	Find volume with $V = l \times w \times h$	**5.MD.C.5.b**
FORWARD 1	Compare the volumes of cylinders and prisms	**8.G.C.9**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

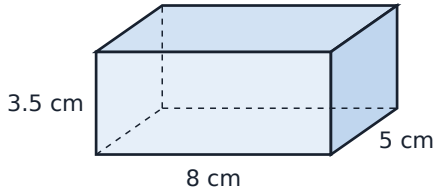
SET 47 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the volume of the right rectangular prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q1/5

1434

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

140 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q1/5

A right triangular prism has triangular bases with a base of 6 m and a height of 4 m. The prism is 10 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q2/5

1435

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

120 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q2/5

The base of a right prism has an area of 24 square inches. The prism is 7.5 inches tall.

What is its volume?

A. 90 cubic inches

B. 180 cubic inches

C. 360 cubic inches

D. 31.5 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q3/5

1436

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 180 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q3/5

True or false?

A cube with $\frac{1}{2}$-foot edges has a volume of $\frac{1}{8}$ cubic foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q4/5

1437

QUESTION ID

S47-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q4/5

True or false?

Doubling the height of a right prism doubles the area of its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q5/5

1438

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It doubles the volume; the base does not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q5/5

SET 47 • 7.G.B.6

BACKWARD 1

Find the volume of a rectangular prism with fractional edges • Grade 6 • Geometry • 6.G.A.2

A right rectangular prism is $2\frac{1}{2}$ feet by 2 feet by 4 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q1/5

1440

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q1/5

A cube has $\frac{3}{4}$-inch edges.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q2/5

1441

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27/64 cubic inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q2/5

A right rectangular prism is $\frac{1}{2}$ m by 3 m by 6 m.

What is its volume?

A. 9.5 cubic meters

B. 4.5 cubic meters

C. 9 cubic meters

D. 18 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q3/5

1442

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A right rectangular prism $1\frac{1}{2}$ cm by 2 cm by 2 cm has a volume of 6 cubic centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q4/5

1443

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A right rectangular prism $\frac{1}{2}$ foot by $\frac{1}{2}$ foot by 4 feet has a volume of 2 cubic feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q5/5

1444

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 1 cubic foot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 7.G.B.6

BACKWARD 2

Find volume with V = l × w × h • Grade 5 • Measurement & Data • 5.MD.C.5.b

A rectangular prism is 7 units long, 4 units wide, and 3 units tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q1/5

1446

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

84 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q1/5

A box is 10 cm long, 6 cm wide, and 5 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q2/5

1447

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

300 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q2/5

A rectangular prism is 9 feet by 2 feet by 3 feet.

What is its volume?

A. 27 cubic feet

B. 14 cubic feet

C. 108 cubic feet

D. 54 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q3/5

1448

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 54 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q4/5

1449

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q4/5

True or false?

A box 5 m by 5 m by 2 m has a volume of 12 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q5/5

1450

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 50 cubic meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 7.G.B.6

FORWARD 1

Compare the volumes of cylinders and prisms • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 4 inches and a height of 5 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q1/5

1452

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

251.2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q1/5

A cylinder's base has an area of 12.56 square centimeters. The cylinder is 10 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q2/5

1453

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

125.6 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q2/5

A cylinder has a radius of 1 m and a height of 3 m.

What is its volume in terms of π?

A. π cubic meters

B. 9π cubic meters

C. 6π cubic meters

D. 3π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q3/5

1454

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q3/5

True or false?

The volume of a cylinder is its base area times its height, just like a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q4/5

1455

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 2 has a volume of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q5/5

1456

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 8π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 7.G.B.6

FORWARD 2

Use volume formulas to solve problems • Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A pyramid has a square base with 6 m sides. It is 10 m tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q1/5

1458

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q1/5

A cone-shaped cup has a radius of 3 cm and holds 30π cubic centimeters.

How tall is the cup?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q2/5

1459

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q2/5

A sphere has a volume of 288π cubic centimeters.

What is its radius?

A. 6 cm

B. 12 cm

C. 4 cm

D. 8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q3/5

1460

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q3/5

True or false?

A pyramid has $\frac{1}{3}$ the volume of a prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q4/5

1461

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q4/5

True or false?

A sphere with a radius of 3 has a volume of 27π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q5/5

1462

QUESTION ID

S47-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 36π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q5/5

SET 48 • 7.G.B.6 • GEOMETRY

Find the surface area of right prisms

MAIN	Grade 7 main question	7.G.B.6
BACKWARD 1	Use a net to find surface area	6.G.A.4
BACKWARD 2	Find the area of a rectangle	4.MD.A.3
FORWARD 1	Use the Pythagorean theorem in three-dimensional figures — near…	8.G.B.7
FORWARD 2	Apply geometric methods to solve design problems	HSG-MG.A.3

SET 48 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the surface area of a right rectangular prism that is 6 cm by 4 cm by 3 cm.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q1/5

1465

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

108 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q1/5

Find the surface area of a cube with 5-inch edges.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q2/5

1466

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

150 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q2/5

A right triangular prism has triangular bases with sides of 3 cm, 4 cm, and 5 cm and a right angle between the 3 cm and 4 cm sides. The prism is 10 cm long.

What is its surface area?

A. 126 square centimeters **B.** 60 square centimeters

C. 132 square centimeters **D.** 120 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q3/5

1467

QUESTION ID

S48-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 132 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q3/5

True or false?

A box 10 in by 2 in by 2 in needs exactly 88 square inches of paper to cover it with no overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q4/5

1468

QUESTION ID

S48-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q4/5

True or false?

A right rectangular prism 2 m by 3 m by 4 m has a surface area of 24 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q5/5

1469

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its surface area is 52 square meters; 24 cubic meters is its volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q5/5

SET 48 • 7.G.B.6

BACKWARD 1

Use a net to find surface area • Grade 6 • Geometry • 6.G.A.4

The net of a cube has 6 squares. Each square is 4 cm by 4 cm.

What is the surface area of the cube?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q1/5

1471

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

96 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q1/5

The net of a rectangular prism has two 5-by-3 rectangles, two 5-by-2 rectangles, and two 3-by-2 rectangles.

What is the surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q2/5

1472

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

62 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q2/5

The net of a square pyramid has a 6-by-6 square and 4 triangles. Each triangle has a base of 6 and a height of 5.

What is the surface area?

A. 186 square units

B. 66 square units

C. 156 square units

D. 96 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q3/5

1473

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 96 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q3/5

True or false?

A net shows all the faces of a solid laid flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q4/5

1474

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q4/5

True or false?

The net of a rectangular prism has 4 rectangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q5/5

1475

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 6 rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 7.G.B.6

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 8 cm by 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q1/5

1477

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

40 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q1/5

A rectangle is 14 inches by 3 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q2/5

1478

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

42 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q2/5

A rectangle is 11 m by 6 m.

What is its area?

A. 34 square meters

B. 66 square meters

C. 17 square meters

D. 60 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q3/5

1479

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 66 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

A rectangle 9 feet by 2 feet has an area of 18 square feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q4/5

1480

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

A rectangle 5 cm by 5 cm has an area of 20 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q5/5

1481

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 25 square centimeters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 7.G.B.6

FORWARD 1

Use the Pythagorean theorem in three-dimensional figures • Grade 8 • Geometry • 8.G.B.7

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A box is 4 units long, 3 units wide, and 12 units tall.

How long is a diagonal of its 4-by-3 base?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q1/5 • NEAREST RELATED

1483

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q1/5 • NEAREST RELATED

A box is 4 units long, 3 units wide, and 12 units tall.

How long is the diagonal from a bottom corner to the opposite top corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q2/5 • NEAREST RELATED

1484

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q2/5 • NEAREST RELATED

A square pyramid has base edges of 6 m and is 4 m tall.

What is the slant height (from the top to the midpoint of a base edge)?

A. 10 m

B. 7 m

C. 5 m

D. 2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q3/5 • NEAREST RELATED

1485

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A rectangle that is 6 by 8 has a diagonal of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q4/5 • NEAREST RELATED

1486

QUESTION ID

S48-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A cube with edges of 1 has a diagonal from one corner to the opposite corner of length 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q5/5 • NEAREST RELATED

1487

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its length is √3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 48 • 7.G.B.6

FORWARD 2

Apply geometric methods to solve design problems • Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a box shaped like a cube with a volume of 64 cubic inches.

What is the edge length, and how much cardboard covers the box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q1/5

1489

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 inches; 96 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q1/5

One gallon of paint covers 350 square feet. The walls of a room total 1,225 square feet.

How many whole gallons should be bought?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q2/5

1490

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 3.5 gallons are needed, so 4 whole gallons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q2/5

Each box has a volume of 24 cubic centimeters.

Which box uses the least cardboard?

A. 1 cm × 4 cm × 6 cm

B. 1 cm × 1 cm × 24 cm

C. 2 cm × 3 cm × 4 cm

D. 1 cm × 2 cm × 12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q3/5

1491

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 cm × 3 cm × 4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q3/5

True or false?

Among rectangular boxes with the same volume, a cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q4/5

1492

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q4/5

True or false?

Two boxes with the same volume always have the same surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q5/5

1493

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q5/5

SET 49 • 7.SP.A.1 • STATISTICS & PROBABILITY

Understand random sampling and representative samples

MAIN	Grade 7 main question	7.SP.A.1
BACKWARD 1	Recognize a statistical question	6.SP.A.1
BACKWARD 2	Report the number of observations	6.SP.B.5.a
FORWARD 1	Summarize survey data in a two-way table — nearest related (no…	8.SP.A.4
FORWARD 2	Recognize surveys, experiments, and observational studies	HSS-IC.B.3

SET 49 • 7.SP.A.1

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know students' favorite lunch.

Which sample is most likely to represent all the students in the school?

A. 20 students in one 7th grade class

B. 20 students on the basketball team

C. The first 20 students in the lunch line

D. 20 students chosen at random from the school list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q1/5

1496

QUESTION ID

S49-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. 20 students chosen at random from the school list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q1/5

A town surveys people at a gym about how often they exercise.

Is the sample likely to represent the whole town? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q2/5

1497

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. People at a gym probably exercise more than people in the whole town, so the sample is biased.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe one way to choose a random sample of 30 students from a school of 600 students.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q3/5

1498

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

Number the students 1 to 600 and use a random number generator to choose 30 different numbers.

GRADING NOTE Any method that gives every student the same chance of being chosen is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q3/5

True or false?

In a random sample, every member of the population has the same chance of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q4/5

1499

QUESTION ID

S49-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q4/5

True or false?

A large sample always represents the population, even if it is not chosen at random.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q5/5

1500

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q5/5

SET 49 • 7.SP.A.1

BACKWARD 1

Recognize a statistical question • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How tall are the students in my class?" a statistical question? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q1/5

1502

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The answers will vary from student to student.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q1/5

Which is a statistical question?

A. How many hours do 7th graders sleep on school nights?

B. What is 5 × 8?

C. How tall is the school flagpole?

D. What day is it today?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q2/5

1503

QUESTION ID

S49-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. How many hours do 7th graders sleep on school nights?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite "How old am I?" as a statistical question.

Question: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q3/5

1504

QUESTION ID

S49-B1-Q3

TYPE

Short answer

CORRECT ANSWER

How old are the students in my school?

GRADING NOTE Any question whose answers vary, such as one about a group's ages, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

"What are the shoe sizes of the players on the team?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q4/5

1505

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

"How many days are in March?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q5/5

1506

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 7.SP.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

How many observations does the dot plot show?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q1/5

1508

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q1/5

A survey asks 25 students how many pets they have. Each student gives one answer.

How many observations are there?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q2/5

1509

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q2/5

A table lists the heights of 18 plants, one height for each plant.

How many observations are there?

A. 1

B. 36

C. 18

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q3/5

1510

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q3/5

True or false?

Each dot on a dot plot shows one observation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q4/5

1511

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q4/5

True or false?

A data set of 12 test scores has 24 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q5/5

1512

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 7.SP.A.1

FORWARD 1

Summarize survey data in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What fraction of the surveyed students play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q1/5 • NEAREST RELATED

1514

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 40/80 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What fraction of all surveyed students have a job and play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q2/5 • NEAREST RELATED

1515

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/20

GRADING NOTE 12/80, or 0.15, is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Among students who have a job, what fraction play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. $\frac{3}{10}$ **B.** $\frac{3}{5}$ **C.** $\frac{2}{5}$ **D.** $\frac{3}{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q3/5 • NEAREST RELATED

1516

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, 56% of the students with no job play a sport.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q4/5 • NEAREST RELATED

1517

QUESTION ID

S49-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, more than half of all the students have a job.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q5/5 • NEAREST RELATED

1518

QUESTION ID

S49-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 49 • 7.SP.A.1

FORWARD 2

Recognize surveys, experiments, and observational studies • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 50 plants to get a new plant food and 50 plants to get none.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q1/5

1520

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

An experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q1/5

A study records the screen time of 200 teens without changing anything they do.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q2/5

1521

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

An observational study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q2/5

Why is random assignment used in an experiment?

A. To avoid collecting data

B. To make the sample larger

C. To make the groups alike, so differences can be linked to the treatment

D. To let people choose their own group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q3/5

1522

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. To make the groups alike, so differences can be linked to the treatment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q3/5

True or false?

A survey asks a sample of people questions in order to estimate facts about a population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q4/5

1523

QUESTION ID

S49-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q4/5

True or false?

An observational study can prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q5/5

1524

QUESTION ID

S49-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use data from a random sample to draw inferences about a population

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find what percent a part is of a whole	**6.RP.A.3.c**
BACKWARD 2	Find a missing value in equivalent ratios	**6.RP.A.3.a**
FORWARD 1	Use a linear model to make a prediction — nearest related (no di…	**8.SP.A.3**
FORWARD 2	Use a margin of error to describe a population estimate	**HSS-IC.B.4**

SET 50 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 50 students, 18 prefer the art club.

Predict how many of the 800 students in the school prefer the art club.

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q1/5

1527

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q1/5

In a random sample of 40 light bulbs, 2 do not work.

Estimate how many bulbs in a shipment of 2,000 do not work.

Bulbs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q2/5

1528

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q2/5

In a random sample of 60 voters, 33 support a plan.

What is the best estimate of the percent of all voters who support it?

A. 45%

B. 33%

C. 60%

D. 55%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q3/5

1529

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 55%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q3/5

True or false?

In a random sample of 25 fish from a lake, 5 are trout. A good estimate is that 20% of the fish in the lake are trout.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q4/5

1530

QUESTION ID

S50-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q4/5

True or false?

In a random sample of 30 homes, 9 have a dog. A good estimate is that 900 of 1,500 homes have a dog.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q5/5

1531

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A good estimate is 450 homes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 1

Find what percent a part is of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

18 out of 50 is what percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q1/5

1533

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q1/5

10 is what percent of 40?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q2/5

1534

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

7 out of 20 is what percent?

A. 35%

B. 7%

C. 20%

D. 70%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q3/5

1535

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 35%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q3/5

True or false?

12 out of 48 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q4/5

1536

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

9 out of 30 is 9%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q5/5

1537

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 out of 30 is 30%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 2

Find a missing value in equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Find the missing value.

2 : 5 = ? : 100

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q1/5

1539

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q1/5

Find the missing value.

$4 : 9 = 28 : ?$

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q2/5

1540

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 6 : 15?

A. 3 : 5

B. 6 : 5

C. 2 : 5

D. 1 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q3/5

1541

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 : 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

5 : 8 = 25 : 40

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q4/5

1542

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$3 : 7 = 9 : 14$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q5/5

1543

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 7.SP.A.2

FORWARD 1

Use a linear model to make a prediction • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A line of fit is y = 2.5x + 40, where x is hours studied and y is a test score.

Predict the score for 8 hours of study.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q1/5 • NEAREST RELATED

1545

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q1/5 • NEAREST RELATED

A line of fit is y = 2.5x + 40, where x is hours studied and y is a test score.

What does 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q2/5 • NEAREST RELATED

1546

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Each extra hour of study goes with a score about 2.5 points higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q2/5 • NEAREST RELATED

A line of fit is y = −3x + 90.

What does it predict for x = 10?

A. 120

B. 93

C. 87

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q3/5 • NEAREST RELATED

1547

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

For the line of fit y = 0.5x + 12, the predicted y-value at x = 0 is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q4/5 • NEAREST RELATED

1548

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

The line of fit $y = 4x + 1$ predicts $y = 41$ when $x = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q5/5 • NEAREST RELATED

1549

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It predicts $y = 37$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 50 • 7.SP.A.2

FORWARD 2

Use a margin of error to describe a population estimate • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample shows 52% support for a plan, with a margin of error of 4%.

What interval of values is likely for the whole population?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q1/5

1551

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

48% to 56%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q1/5

A random sample has a mean of 6.8 hours with a margin of error of 0.5 hour.

What interval of values is likely for the population mean?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q2/5

1552

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6.3 to 7.3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q2/5

As a random sample gets larger, what usually happens to the margin of error?

A. It gets larger.

B. It becomes 0 once the sample is over 10.

C. It gets smaller.

D. It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q3/5

1553

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. It gets smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

A margin of error describes how far a sample estimate is likely to be from the population value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q4/5

1554

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

A poll result of 52% with a 4% margin of error proves that more than half of the population supports the plan.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q5/5

1555

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The interval 48% to 56% includes values below 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use several random samples to judge how much estimates vary

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find the mean of a data set	**6.SP.B.5.c**
BACKWARD 2	Find the range as a measure of variation	**6.SP.A.3**
FORWARD 1	Describe the association shown in a scatter plot — nearest relate…	**8.SP.A.1**
FORWARD 2	Decide whether a result is consistent with a model by simulation	**HSS-IC.A.2**

SET 51 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

Four random samples of 20 students give these mean hours of sleep: 7.8, 8.2, 8.0, and 8.4.

What is a good estimate of the mean for all the students? Use the mean of the samples.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q1/5

1558

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

8.1 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q1/5

Four random samples give mean estimates of 7.8, 8.2, 8.0, and 8.4 hours.

What is the range of the estimates?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q2/5

1559

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.6 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q2/5

Three random samples of 50 people show that 22%, 26%, and 24% prefer blue.

Which conclusion is best?

A. Exactly 22% of the population prefers blue.

B. Exactly 26% of the population prefers blue.

C. The samples are useless because they differ.

D. About 24% of the population prefers blue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q3/5

1560

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. About 24% of the population prefers blue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q3/5

True or false?

Different random samples from the same population usually give somewhat different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q4/5

1561

QUESTION ID

S51-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q4/5

True or false?

Small random samples usually give estimates that vary less than large random samples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q5/5

1562

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 7, 9, 10

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q1/5

1564

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q1/5

Find the mean.

12, 15, 18

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q2/5

1565

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q2/5

What is the mean of 3, 3, 6, and 8?

A. 5

B. 6

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q3/5

1566

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q3/5

True or false?

The mean of 2, 4, 6, and 8 is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q4/5

1567

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q4/5

True or false?

The mean of 10, 20, and 60 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q5/5

1568

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The mean is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 2

Find the range as a measure of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the range.

12, 18, 9, 15

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q1/5

1570

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q1/5

Find the range.

3.5, 7, 4.25, 6

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q2/5

1571

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which data set has the greatest range?

A. 1, 3, 2

B. 7, 9, 8

C. 4, 6, 5

D. 2, 10, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q3/5

1572

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2, 10, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q3/5

True or false?

The range of 20, 25, and 30 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q4/5

1573

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q4/5

True or false?

A data set whose values are all the same has a range of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q5/5

1574

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its range is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 7.SP.A.2

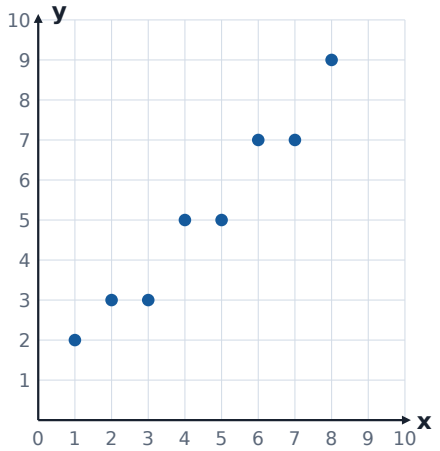
FORWARD 1

Describe the association shown in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

What kind of association does the scatter plot show?

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q1/5 • NEAREST RELATED

1576

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot shows hours of practice (x) and free throws made (y). The points rise from left to right in a roughly straight pattern.

Describe the association.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q2/5 • NEAREST RELATED

1577

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q2/5 • NEAREST RELATED

A scatter plot shows the age of a car (x) and its value (y).

What association is most likely?

A. Negative association

B. Every point on one line

C. No association

D. Positive association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q3/5 • NEAREST RELATED

1578

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Negative association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a scatter plot with no association, the points show no clear trend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q4/5 • NEAREST RELATED

1579

QUESTION ID

S51-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Points that fall from left to right in a scatter plot show a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q5/5 • NEAREST RELATED

1580

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 51 • 7.SP.A.2

FORWARD 2

Decide whether a result is consistent with a model by simulation • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A coin is said to be fair. In 100 simulated sets of 20 fair flips, 15 or more heads happened only 2 times. A real coin gives 15 heads in 20 flips.

Is the fair-coin model believable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q1/5

1582

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Probably not. A result that extreme happens only about 2% of the time with a fair coin, so the result casts doubt on the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q1/5

Simulations of a spinner model usually give 3 to 7 reds in 20 spins. A real spinner gives 5 reds in 20 spins.

What does this show?

A. The spinner is proven unfair.

B. The result is impossible.

C. The result is consistent with the model.

D. The model is proven wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q2/5

1583

QUESTION ID

S51-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The result is consistent with the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q2/5

In 100 simulations of a model, a certain result happened 40 times.

Is that result unusual for the model? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q3/5

1584

QUESTION ID

S51-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No. It happened about 40% of the time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q3/5

True or false?

A result that rarely happens in simulations of a model suggests that the model may be wrong.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q4/5

1585

QUESTION ID

S51-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q4/5

True or false?

A result that happens even once in 1,000 simulations is a common result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q5/5

1586

QUESTION ID

S51-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q5/5

SET 52 • 7.SP.B.3 • STATISTICS & PROBABILITY

Describe how much two distributions overlap using the difference in means as a multiple of the MAD

MAIN	Grade 7 main question	**7.SP.B.3**
BACKWARD 1	Find the mean absolute deviation (MAD)	**6.SP.B.5.c**
BACKWARD 2	Describe the shape of a distribution	**6.SP.A.2**
FORWARD 1	Compare groups using relative frequencies in a two-way table —…	**8.SP.A.4**
FORWARD 2	Compare the center and spread of two data sets	**HSS-ID.A.2**

SET 52 • 7.SP.B.3

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.3

Team A's heights have a mean of 160 cm. Team B's heights have a mean of 172 cm. Both teams have a MAD of 4 cm.

The difference in the means is how many MADs?

MADs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q1/5

1589

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q1/5

Class 1 has a mean quiz score of 82. Class 2 has a mean of 76. Each class has a MAD of 3.

Write the difference in means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q2/5

1590

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

2 MADs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q2/5

Two data sets have means of 50 and 54. Each has a MAD of 8.

How much do the distributions overlap?

A. A lot, because the difference is only half of a MAD

B. Not at all, because the difference is 4 MADs

C. Very little, because the difference is 8 MADs

D. Not at all, because the means are different

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q3/5

1591

QUESTION ID

S52-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A lot, because the difference is only half of a MAD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q3/5

True or false?

When the difference in means is several MADs, the two distributions overlap very little.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q4/5

1592

QUESTION ID

S52-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q4/5

True or false?

Two data sets have means of 30 and 40 and a MAD of 10 each. The difference in means is 10 MADs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q5/5

1593

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 1 MAD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 1

Find the mean absolute deviation (MAD) • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the MAD.

2, 4, 6, 8

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q1/5

1595

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q1/5

Find the MAD.

5, 5, 9, 13

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q2/5

1596

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the MAD of 10, 10, 10, 10?

A. 0

B. 40

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q3/5

1597

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

The MAD of 1, 3, 5, 7 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q4/5

1598

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

The MAD of 0 and 10 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q5/5

1599

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The MAD is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 2

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2